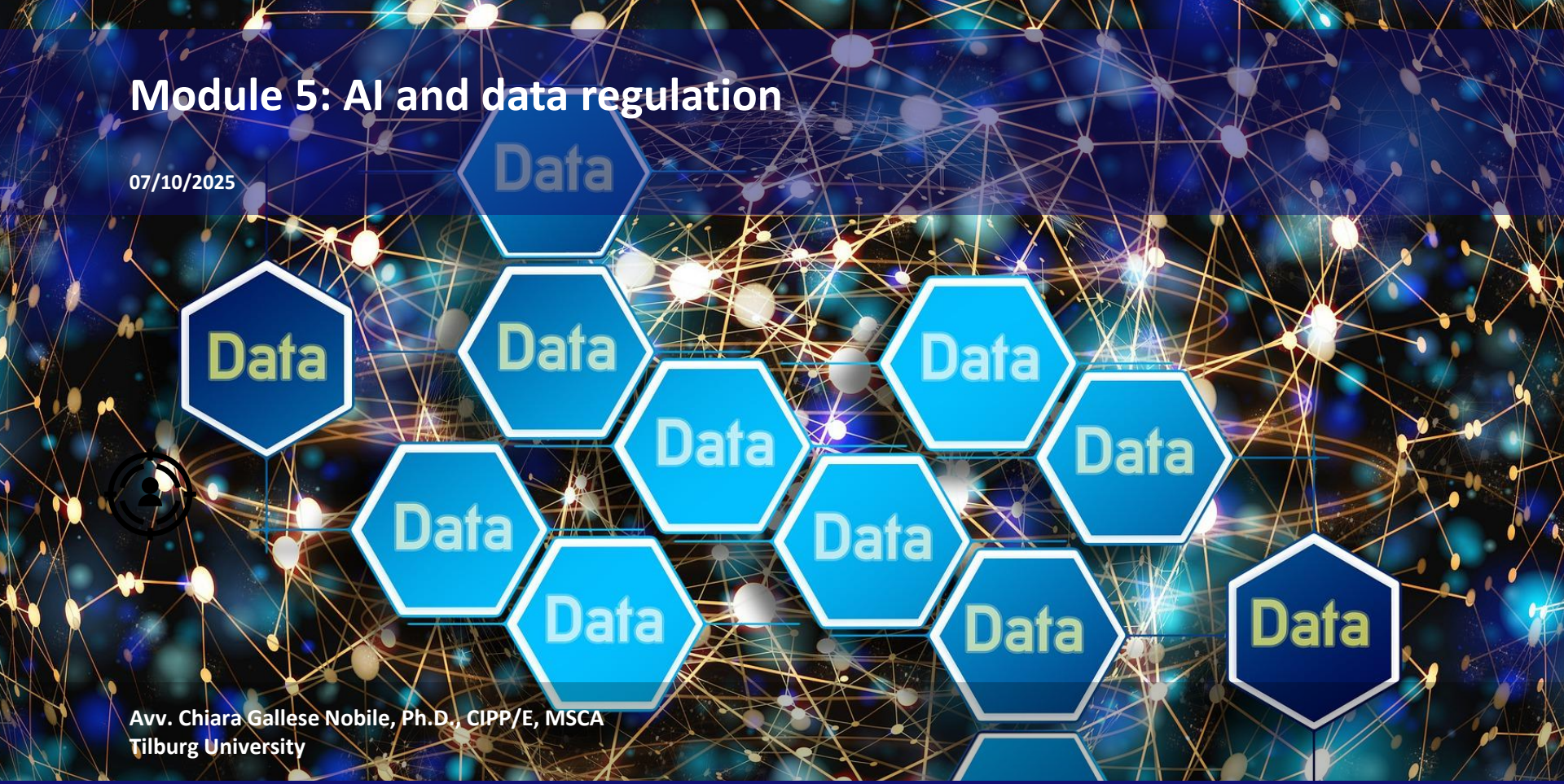
Module 5: AI and data regulation

07/10/2025



Avv. Chiara Gallese Nobile, Ph.D., CIPP/E, MSCA
Tilburg University

Outline

- Introduction: data collection for machine learning
1) Basic principles of Data Protection
2) Basic principles of AI Auditing
3) Basic principles of AI Ethics
4) The new AI Act proposal
5) Applying ethical principles to data sets: practical examples
- Conclusion
- Q&A

INTRODUCTION

WHY THIS COURSE?

You will be able to:

- Understand data collection practices
- Recognize applicable laws and regulations
- Recognize ethical principles related to AI
- Apply theoretical principles in practice
- Avoid sanctions and reputational damage

RECENT LAWSUIT & SANCTION CASES

IBM

ClearviewAI

ChatGPT

What did they do wrong?

WRONG DATA COLLECTION!

Data Scraping

Scraped the web collecting images, code and texts without consent

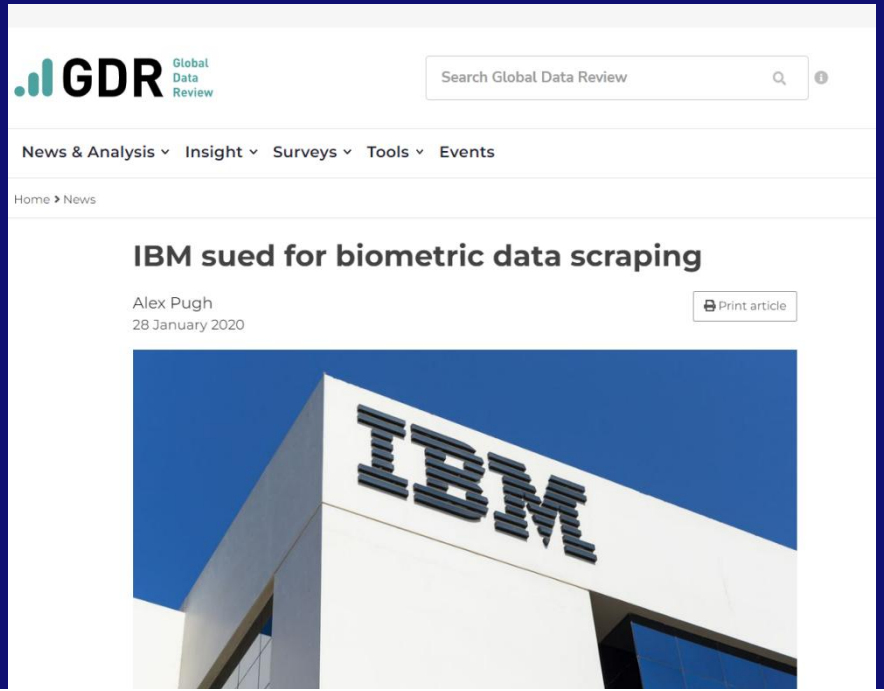
Information

Did not provide information about their practices

Data use

Used copyrighted materials and personal data without authorization

IBM



Allegedly scraped the web to collect random people images

CLEARVIEW AI

Selfie scraping Clearview AI hit with another €20M ban order in Europe

Natasha Lomas @riptari / 5:03 PM GMT+2 • July 13, 2022 Comment

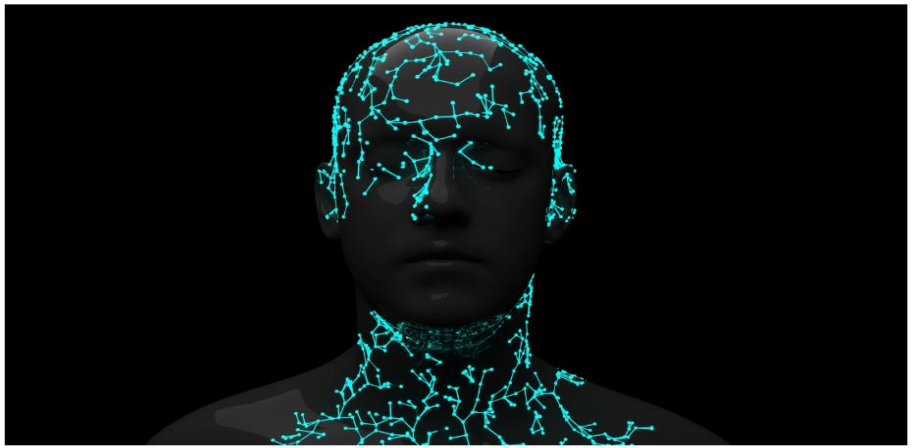
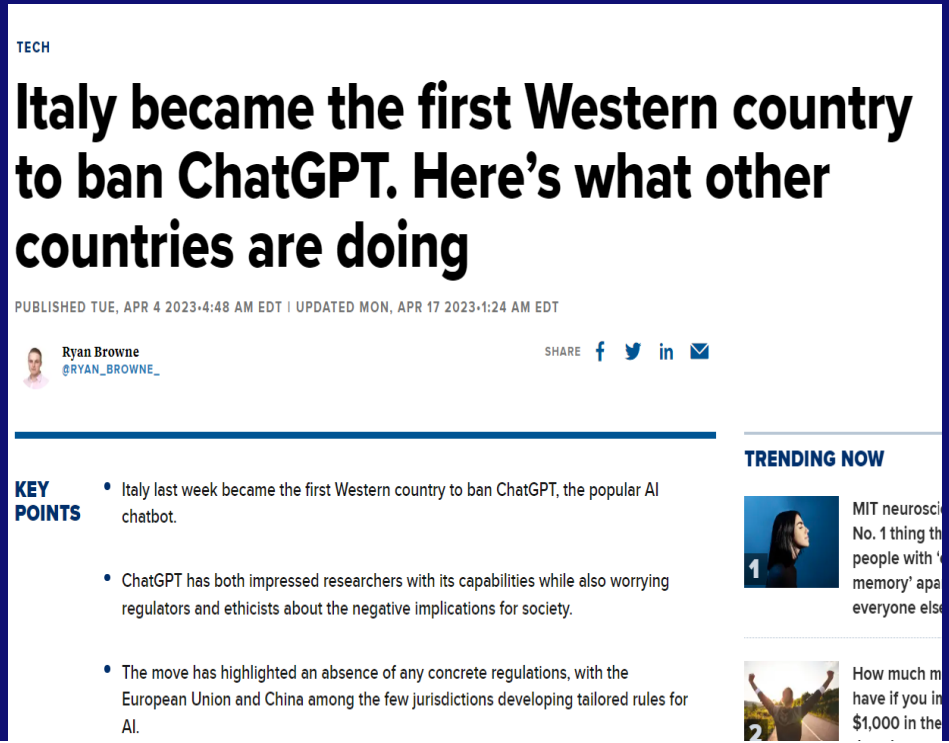


Image Credits: Design Cells / Getty Images

Clearview AI has been hit with another sanction for breaching European privacy rules.

Used innocent people faces to train their AI system aimed at recognizing criminals in public spaces by Law Enforcment Authorities

OPEN AI



TECH

Italy became the first Western country to ban ChatGPT. Here's what other countries are doing

PUBLISHED TUE, APR 4 2023•4:48 AM EDT | UPDATED MON, APR 17 2023•1:24 AM EDT

Ryan Browne
@RYAN_BROWNE_

SHARE

KEY POINTS

- Italy last week became the first Western country to ban ChatGPT, the popular AI chatbot.
- ChatGPT has both impressed researchers with its capabilities while also worrying regulators and ethicists about the negative implications for society.
- The move has highlighted an absence of any concrete regulations, with the European Union and China among the few jurisdictions developing tailored rules for AI.

TRENDING NOW

1 MIT neurosci… No. 1 thing th… people with '… memory' apa… everyone else…

2 How much m… have if you in… $1,000 in the…

Scraped the web to indiscriminately collect personal data and copyrighted material

…AND MORE

LEADERSHIP INSIGHTS

In generative AI legal Wild West, the courtroom battles are just getting started

PUBLISHED MON, APR 3 2023•10:56 AM EDT | UPDATED MON, APR 3 2023•11:29 AM EDT



Ellen Sheng
@ELLENSHENG

SHARE

KEY POINTS

- As companies including Microsoft, Google and OpenAI launch generative AI to the general public, lawsuits are piling up from creative industries about copyrighted work co-opted or used by AI.

RELATED



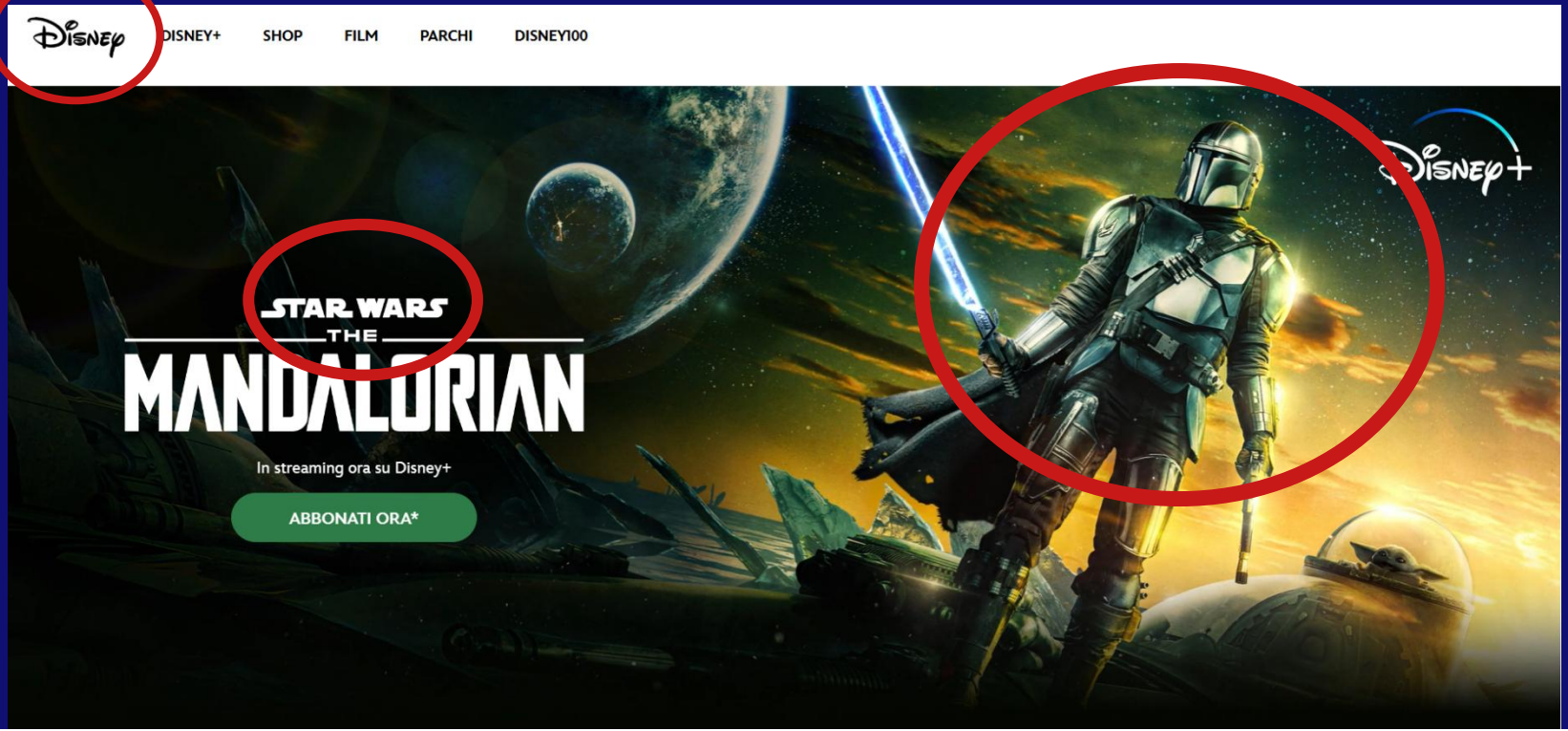
A me
CEO
from

TU/e

WHY COULDN'T THEY COLLECT PUBLICLY AVAILABLE DATA?

1. **COPYRIGHT LAWS**. «**Publicly available**» does not equal to «**public domain**». Trademarks, logos, databases, and copyrighted materials are still protected by law even if they are accessible to the public on a website.
2. **DATA PROTECTION LAWS**. **Personal data** of living persons are still protected by privacy laws even if they are available online.
3. **WEB SCRAPING LAWS**. In some countries, such as Europe, **web scaping** is regulated.
4. **CONTRACT LAWS**. Each website has different **contractual rules**.
5. **ETHICS**. There are some **ethics rules** that prevent from stealing images without consent, whitout information, and for detrimental uses.

COPYRIGHT LAWS



DATA PROTECTION LAWS

1. CONVENTION 108+ (55 Countries)
2. GDPR (the whole EU)
3. NATIONAL LAWS (each country)

NAMES AND OTHER INFORMATION IN WEBSITES ARE STILL PROTECTED BY DATA PROTECTION LAWS!

THE DATABASE DIRECTIVE

NOT AI TRAINING

In the case between **Ryanair** and the websites **Volagratis and Lastminute**, scraping activity **to provide comparison in tickets prices** was deemed lawful by the Court of First Instance, the Court of Appeal of Milan, and the Civil Cassation insofar as copyright protection of the information contained on Ryanair's site was **excluded**, since "the organization of the data is inspired by management purposes and performs functions of information on the products of the company that manages the site, **without an appreciable degree of creative intellectual contribution**". The sui generis right was excluded as well, because, according to the judge, the fact that Ryanair granted the right to extract its data for a 'symbolic' price of €100.00 would demonstrate not the wish to protect such data, but rather the aim of preventing third parties from marketing its flights.

WEB SCRAPING LAWS 1

NOT AI TRAINING

EU LAW: Directive (EU) 2019/790 on copyright and related rights in the Digital Single Market ('DSM Directive'), Article 4, provides **an exception from copyright** ONLY for "**reproductions and extractions** of lawfully accessible works and other subject matter **for the purposes of TEXT AND DATA MINING***", **except** where "the use of works and other subject matter referred to in that paragraph **has not been expressly reserved** by their rightholders in an appropriate manner, such as machine-readable means in the case of content made publicly available online".

* automated computational **analysis of information** in digital form

WEB SCRAPING LAWS 2

US LAW: In 2021, the Supreme Court decision **Van Buren v. United States** stated that scraping publicly available internet information isn't criminalized by the federal computer crime law.
In the long legal battle between **LinkedIn Corp. and hiQ Labs Inc.**, a California District Court in 2017 and Ninth Circuit panel in 2019 first ruled in favor of the latter.

See the case description and links to relevant documents in the article on the National Law Review at the following link:
https://www.natlawreview.com/article/hiq-and-linkedin-reach-proposed-settlement-landmark-scraping-case

LINKEDIN CORP. v. HIQ LABS INC.

HiQ Labs was a small company that began scraping public profile data and data restricted to registered users from LinkedIn users in order to create a **business analytic service**. HiQ Labs filed a declaratory judgment request for a preliminary injunction after receiving a cease-and-desist letter from LinkedIn demanding that hiQ discontinue its scraping activities. The case went to the **Supreme Court**, which **granted LinkedIn's petition** for certiorari and **returned the matter** to the Ninth Circuit for further consideration. In April 2022, **the Ninth Circuit upheld** the lower court's preliminary injunction **in favor of hiQ**, however, in October of the same year, a **California District Court ruled in favor of LinkedIn**. This most recent ruling is noteworthy since it primarily addresses **contract liability** for violating a website's user agreement and its restrictions on scraping, as well as making fraudulent accounts.

RYANAIR DAC v. BOOKING HOLDINGS INC. 1

In **Ryanair DAC v. Booking Holdings Inc.** the famous travel company allegedly acquired information by scraping the ticketing portion of the Ryanair website. According to Ryanair, the action of the defendant violated the Computer Fraud and Abuse Act (CFAA).

The Court held that ``In order for the CFAA's "without authorization" and "exceeds authorized access" elements to apply, some sort of **authentication mechanism** (e.g., the use of usernames and passwords) must be employed to limit access to the website. If the information on the website is **publicly available** without requiring users to authenticate themselves, a violation of the terms of use or the defiance of a cease-and-desist letter will not give rise to liability under the CFAA".

RYANAIR DAC v. BOOKING HOLDINGS INC. 2

However, the Court also noted that ``Ryanair also alleges that the defendants engaged in screen scraping in violation of the terms of use of the Ryanair website and that the defendants continued to do so even after Ryanair sent each defendant a cease-and-desist letter. Although those allegations would be insufficient to establish liability under the CFAA if the contents of the myRyanair portion of the website were accessible to the public without authentication, **courts have found cease-and-desist letters to withdraw authorization to access** a protected portion of a website when an authentication mechanism protected access to that portion of the website".

CONTRACT LAWS

1. EU LAW: contract laws differ depending on the single Member States' law.
2. US Case law: see the cases mentioned in the previous slides
3. India Law: electronic contracts are valid and enforceable and violating a contract that prohibits data scraping could be a violation of contract law. It also might violate section 43 of the Information Technology Act, enacted in 2000, which prohibits unauthorized access to a computer resource or extracting data from a computer resource.

The general rule is that web scraping **can be limited by a contractual clause.** However, as we have seen in the Italian Ryanair case, exceptions apply for specific purposes.

PUBLIC DOMAIN DATA

Some digital objects or data are not covered by copyright laws, such as:

- Mere facts
- Works whose copyright expired
- Works whose license allows its use as public domain data (CC0)
- Works that have no creative intellectual content

ILLICIT CONTENT

Sometimes, we can find online some illicit content, such as stolen data, pornographic materials involving minors of non-consenting individuals, information about terrorist activities, and such.

When performing web scraping to train AI system, it is extremely important to exclude this material, as the **mere possession might be considered a criminal offence**.

ETHICS

EU GUIDELINES: AI HLEG Trustworthy AI Guidelines

INTERNATIONAL GUIDELINES:

Universal Guidelines for AI (2018)

OECD AI Principles (2019)

G20 AI Guidelines (2019)

UNESCO Recommendation on AI Ethics (2021)

1. BASIC PRINCIPLES OF DATA PROTECTION

SOURCES OF LAW

GDPR is not the only law applicable to data protection. There are also international conventions, national laws, Constitution, EU treaty, and other sources of law. Legal instruments are arranged **hierarchically.**

SOURCES OF LAW



…implemented differently in each Member State

MATERIAL AND TERRITORIAL SCOPE

GOALS:

Protecting personal data of natural persons
Facilitating the exchange of data within the EU

GDPR DOES NOT APPLY TO:

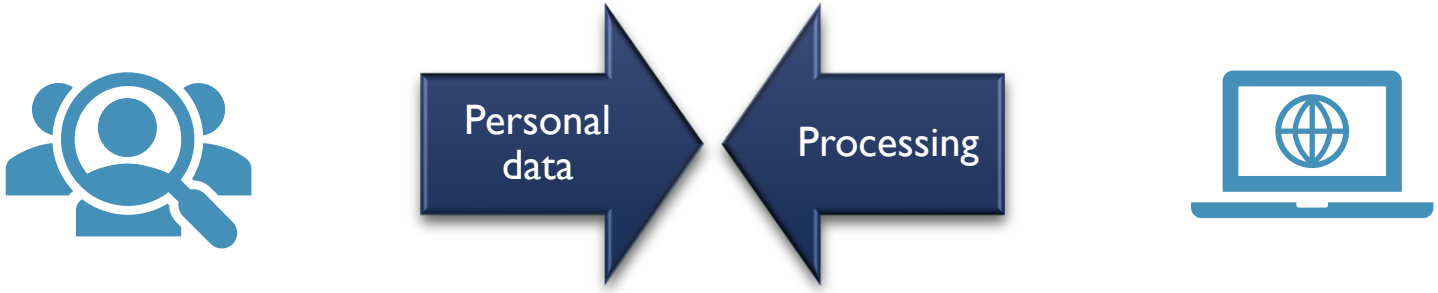
- **Personal** or household activity
- Competent authorities for **criminal offences** or **criminal penalties**
- **EU bodies, offices, agencies**

GDPR APPLIES TO:

- Data **Controller or processor** established in the EU
- Activities **targeting** of subjects located in the EU
- Competent authorities for purposes **other than** law enforcement

(Data Subject's nationality does not matter)

GDPR BASIC PRINCIPLES - DEFINITIONS



Regardless of the format
(mathematical form too)

Data directly or indirectly related to a natural person, such:

- Biometric data (photos, MRI scans, voice, and video recording)
- Genomic expression information
- Profiling (any habits, personality tests, work assessment, preferences)
- Identifiers (nickname, cookies, browser settings)
- Location data (GPS, ZIP code)
- Opinions (religious believes, political opinion)
- Feelings (burnout information, mood information)
- Income, debt and borrower reliability information

DATA COLLECTION IS A FORM OF DATA PROCESSING

- Web scraping
- Text mining
- Receiving data from third parties
- Performing interviews, focus groups, surveys
- Making experiments and clinical analysis
- Analyzing samples
- Online tracking
- GPS monitoring
- Collecting data from IoT

Data processing includes:

- Collecting and Recording data
- Storing, uploading and archiving data
- Modifying, Analyzing and Deleting data
- Using data to create something else (e.g., images, tables)
- Sharing, disseminating, or making data available otherwise
- Organizing or structuring data
- Reading or looking at the data
- Using data to communicate (e.g., sending emails)
- Changing the format or the size of data
- Using data to train a model
- Accessing data through VPN

DATA PROCESSING EXAMPLES IN MACHINE LEARNING

- Collecting data
- Analyzing data
- Putting data into a data set
- Preprocessing data
- Feeding data to train the model

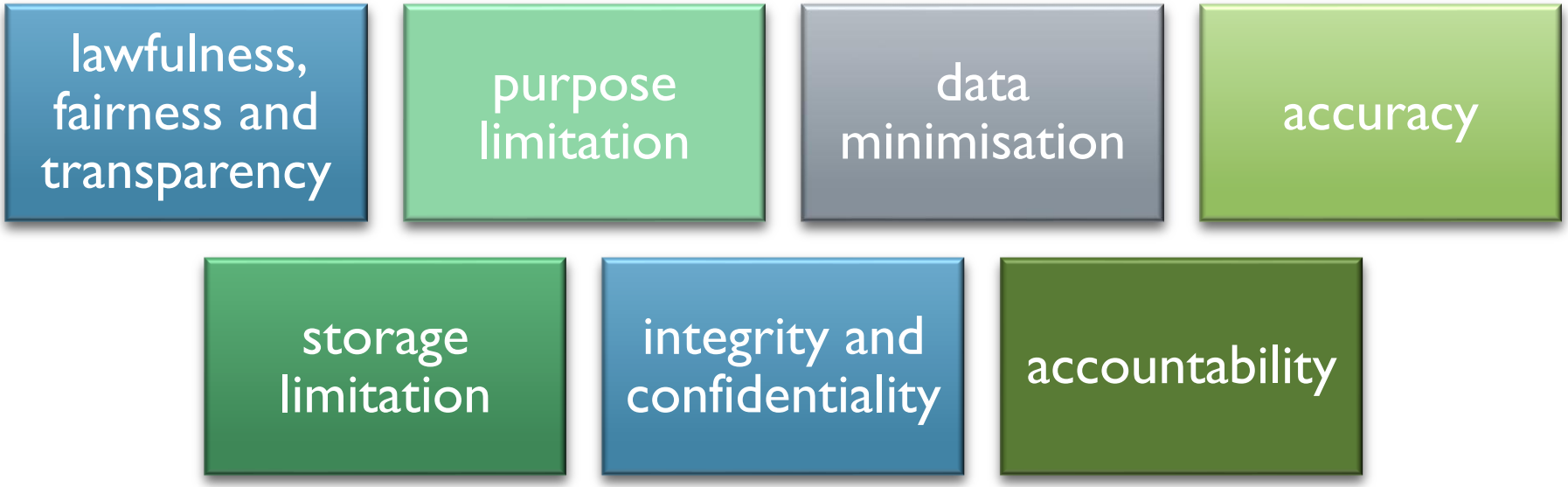
The Machine Learning Life Cycle



1. **Define Project Objectives**
   - ☐ Specify business problem
   - ☐ Acquire subject matter expertise
   - ☐ Define unit of analysis and prediction target
   - ☐ Prioritize modeling criteria
   - ☐ Consider risks and success criteria
   - ☐ Decide whether to continue
2. **Acquire & Explore Data**
   - ☐ Find appropriate data
   - ☐ Merge data into single table
   - ☐ Conduct exploratory data analysis
   - ☐ Find and remove any target leakage
   - ☐ Feature engineering
3. **Model Data**
   - ☐ Variable selection
   - ☐ Build candidate models
   - ☐ Model validation and selection
4. **Interpret & Communicate**
   - ☐ Interpret model
   - ☐ Communicate model insights
5. **Implement, Document & Maintain**
   - ☐ Set up batch or API prediction system
   - ☐ Document modeling process for reproducibility
   - ☐ Create model monitoring and maintenance plan

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GDPR BASIC PRINCIPLES – BASIC CONCEPTS



LAWFULNESS

Legal basis for processing

Respect of GDPR

Respect of other relevant laws and regulations (national and international)

LAWFULNESS IN MACHINE LEARNING

- Make sure you can collect and use the data
- Respect the license of the data set
- Inform the data subjects about data processing, when needed
- Sign the appropriate agreement, when needed
- Discard the data you don't need
- Keep the data safe and secure
- Delete the data when you do not need it anymore
- Document GDPR compliance

FAIRNESS

Consider how it affects the interests of data subject

Consider what the data subject expects

Do not discriminate

FAIRNESS IN MACHINE LEARNING

- Perform the legitimate interest assessment (LIA)
- Perform the risk assessment
- Perform the transfer impact assessment (TIA)
- Perform the re-use assessment
- Perform the data protection impact assessment (DPIA)
- Do not use data to harm data subjects

TRANSPARENCY

Record of activities

Privacy notice/policy

Right to access

PURPOSE LIMITATION

Personal data must be used only for lawful purposes

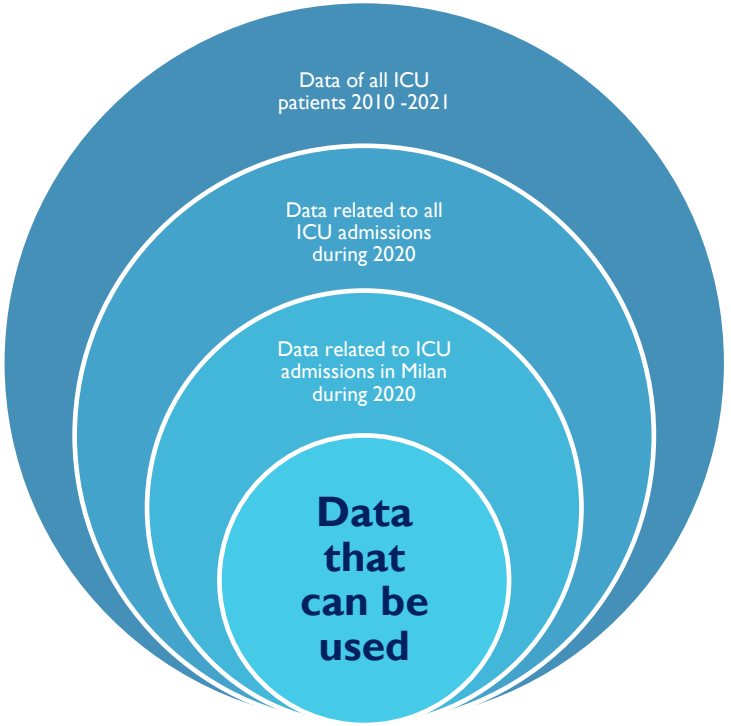
Controller must communicate all purposes to data subjects

Further purposes must be compatible with the original ones



DATA MINIMIZATION

- Only data **strictly necessary** for **each purpose** can be collected, used and shared
- It is not possible to collect data «just in case» they may be useful in the future
- While choosing data, other laws and code of conduct must be considered



Do you need to analyze COVID-19 ICU admissions in Milan during 2020?

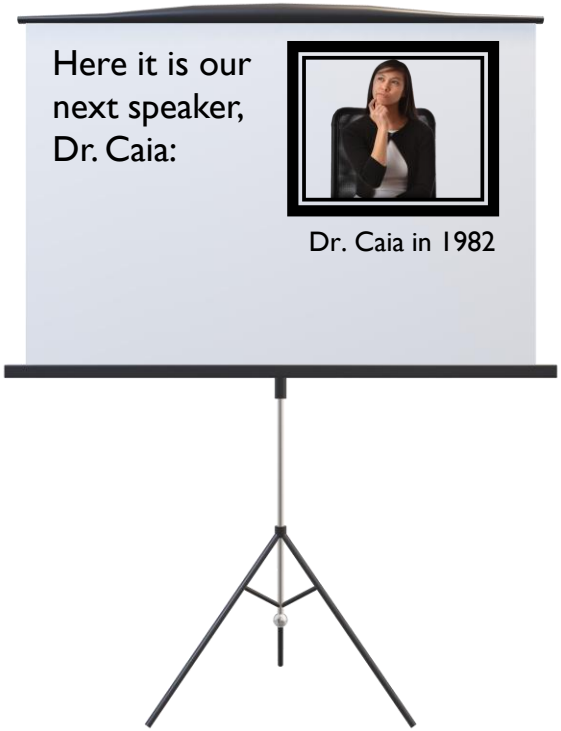
ACCURACY

Personal data must be correct

Personal data must be updated

Data that are not correct or updated must be deleted

2021 Conference





Dr. Caia in 2021

STORAGE LIMITATION

Personal data must be kept only for the time **strictly necessary** for each purpose

After the retention time has expired, data must be deleted

If data must be kept longer for one purpose, data still cannot be used for different purposes

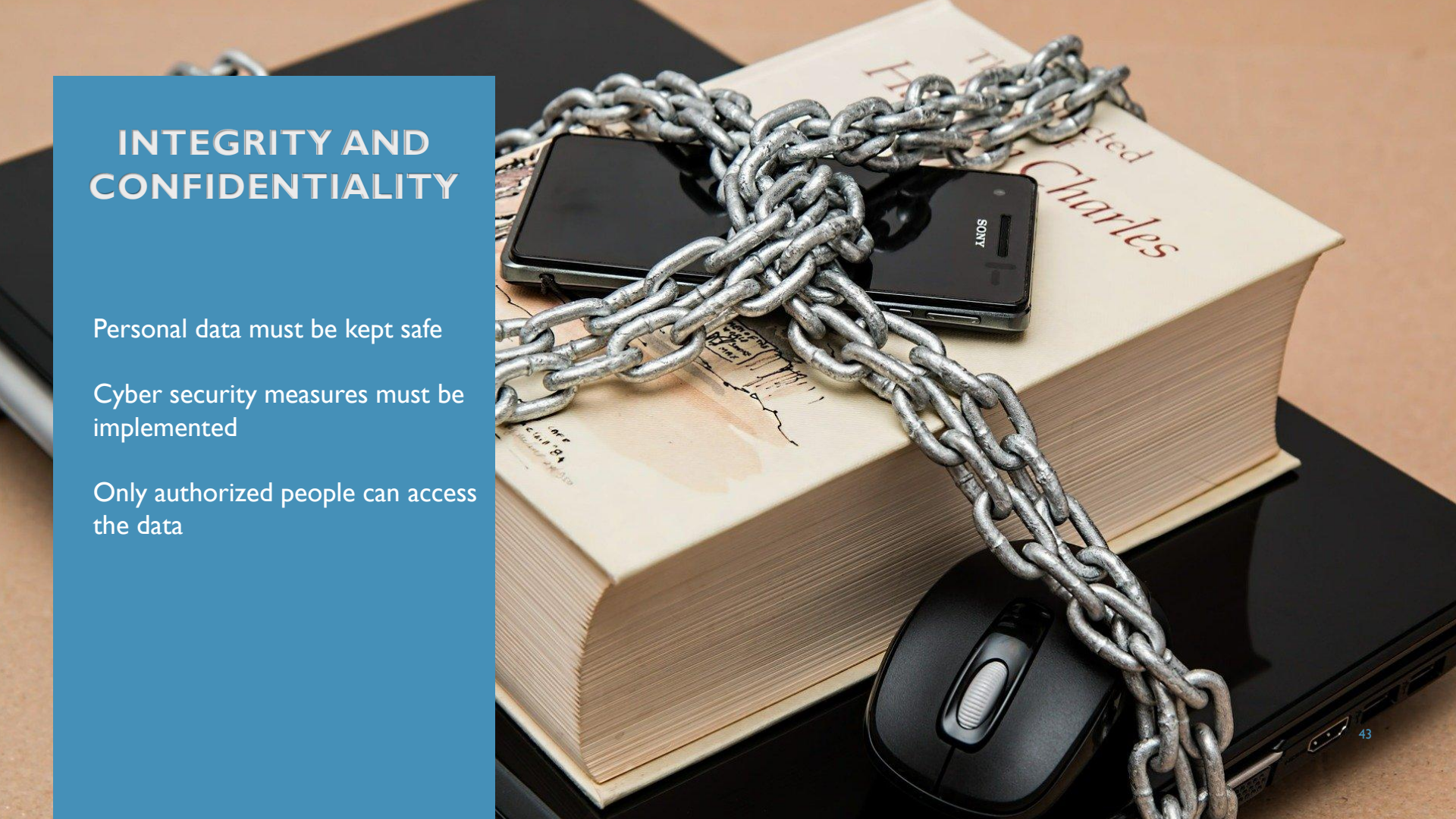


INTEGRITY AND CONFIDENTIALITY

Personal data must be kept safe

Cyber security measures must be implemented

Only authorized people can access the data



ACCOUNTABILITY

Controller is responsible for compliance

There are no minimum requirements, Controller must assess the risks

Controller must demonstrate that everything was done in compliance with GDPR



LEGAL BASIS



Law, regulations, decrees, ordonnances, Court orders



Contract



Consent



Legitimate interest



Public interest



Protecting vital interests

LED Directive

LEGAL BASIS IN AI

- LEGITIMATE INTEREST (research, further use of public data sets)
- CONSENT
- PUBLIC TASK (train models for law enforcement)

In clinical trials, see: https://ec.europa.eu/health/sites/default/files/files/documents/qa_clinicaltrials_gdpr_en.pdf

CAN YOU REUSE SOMEONE ELSE'S DATA SET?

It depends on the specific circumstances. You must assess if the lawful conditions are met:

- Do the other party have the right to give you their data? If not, it is illegal to use the data set
- Do you need an agreement to transfer the data?
- What is your legal basis?

CONSENT

Different from ethical Informed consent

Must be **freely given**, **specific**, **informed** and **unambiguous**

For special categories of data, for transfer abroad, and for automated decision making and profiling, it must be also **explicit**

WHITDRAWAL OF CONSENT

- Data subject has the right to withdraw consent at any time. If it happens, the data processing must be stopped.
- The withdrawal is valid only for future processing, and what has been done before remains lawful.

SPECIAL CATEGORIES OF DATA



Health



Social security number



Ethnicity



Sexual orientation or gender identity



Religious or philosophical beliefs



Political opinions or Trade Union



Genetic



Criminal conviction



Biometric

HEALTH DATA

Physiological measurements (blood pressure, blood analysis, muscles performance)

Psychological state of mind (burnout, mood)

Neurodiversity information (ADHD, Asperger, Dyscalculia, Dyslexia, Non-verbal disorder, etc.)

WHY ARE SPECIAL DATA DIFFERENT FROM OTHER TYPES OF DATA?

- If made public, the consequences may be severe



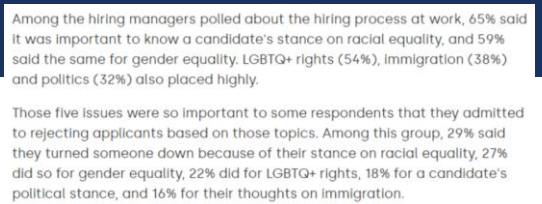
'I didn't know why I'd been declined, so I asked for my medical notes'

Two people with a mental health history describe difficulties of seeking insurance cover

Exclusive: People with mental illnesses refused access to insurance



Caste Surnames of Candidates Should Not be Revealed at UPSC Interview Stage to Decrease Discrimination: Govt Report



Among the hiring managers polled about the hiring process at work, 65% said it was important to know a candidate's stance on racial equality, and 59% said the same for gender equality. LGBTQ+ rights (54%), immigration (38%) and politics (32%) also placed highly.

Those five issues were so important to some respondents that they admitted to rejecting applicants based on those topics. Among this group, 29% said they turned someone down because of their stance on racial equality, 27% did so for gender equality, 22% did for LGBTQ+ rights, 18% for a candidate's political stance, and 16% for their thoughts on immigration.

Personality tests used in the hiring process have consistent patterns of correlations with sociodemographics

Elleman, Lorien G., et al. "The personality of US states: Stability from 1999 to 2015." *Journal of Research in Personality* 72 (2018): 64-72.

DATA CONTROLLER



Data Controller

Legal person, public authority, agency or other body which **determines the purposes and means** of the processing of personal data



Joint Data controllers

Two or more legal persons, public authorities, agencies or other bodies who share and decide the purposes and means of processing

WHEN ARE YOU A DATA CONTROLLER?

- When you collect data
- When you receive data, and you use it for your own purposes
- When you are a processor, and you use data against the Controller's instructions

PROCESSOR



Natural or legal person, public authority, agency or other body which processes personal data **on behalf** of the Data Controller



Provides sufficient **guarantees** to implement appropriate technical and organisational measures



Follows instructions, cooperates, gives notice of Data breaches



Returns or **deletes** Data

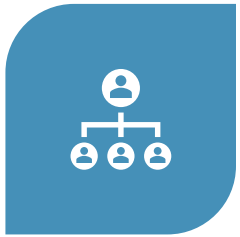
WHEN ARE YOU A DATA PROCESSOR?

- When you are paid to perform analysis on data or to train a model for someone else
- When you help someone for free and you follow their instructions
- When you store data for someone else

SUB-PROCESSOR



NATURAL OR LEGAL PERSON, PUBLIC AUTHORITY, AGENCY OR OTHER BODY WHICH PROCESSES PERSONAL DATA ON BEHALF OF THE CONTROLLER AFTER BEING APPOINTED BY THE PROCESSOR



FOLLOWS INSTRUCTIONS AND COOPERATES, GIVES NOTICE OF DATA BREACHES



RETURNS OR DELETE DATA



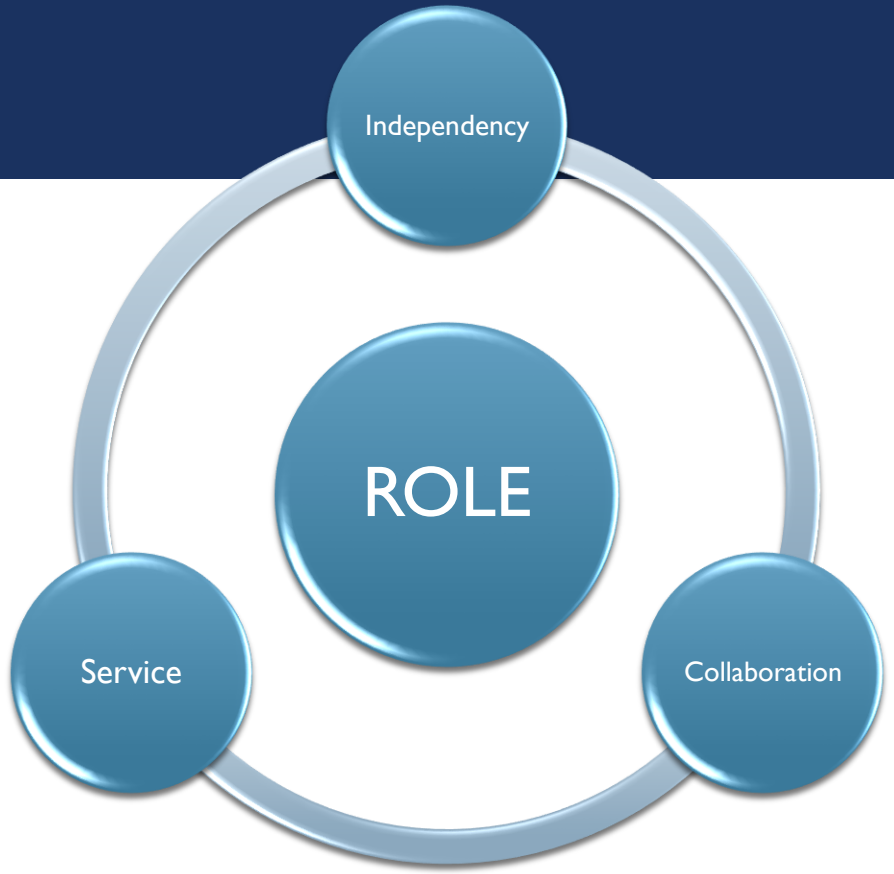
WRITTEN AUTHORIZATION BY THE DATA CONTROLLER, WHO CAN OBJECT

DATA AGREEMENTS

- They are not independent agreements, there has to be an underlying contract, i.e. a reason why data are transfered between Parties

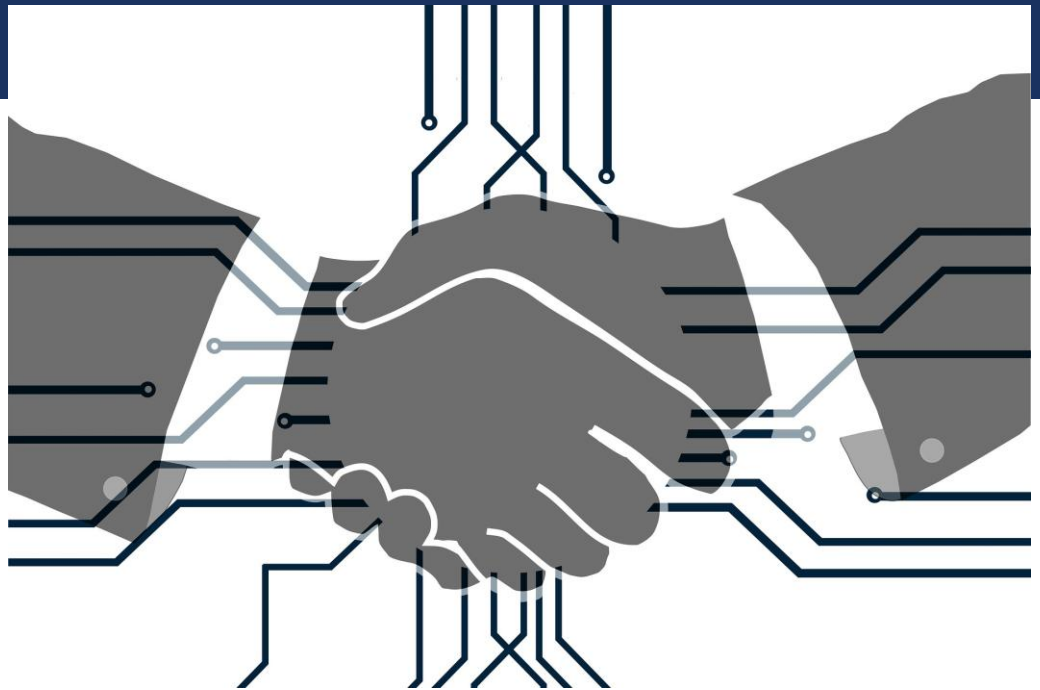


PRIVACY ROLES



DATA TRANSFER/SHARING AGREEMENT

- Between two autonomous Controllers
- When each party determines the purpose of the processing separately
- There must be a legal basis that allows the transfer
- Privacy notice pursuant to art. 14 should be provided at the first contact



JOINT CONTROLLERSHIP AGREEMENT

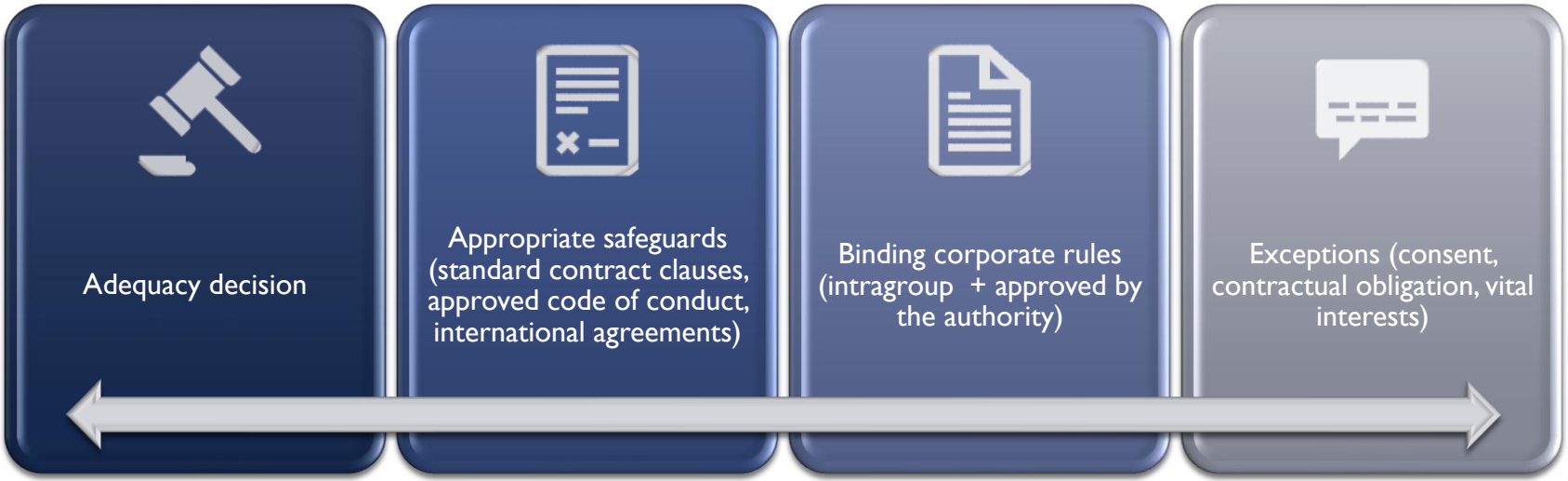
- Between two joint Controllers
- It determines in a transparent manner their respective responsibilities
- It duly reflects the respective roles and relationships of the joint controllers vis-à-vis the data subjects
- The essence of the arrangement shall be made available to the data subject



DATA PROCESSING AGREEMENT

- Between Data Controller(s) and Processor(s)
- A contract or other legal act binding the processor to the controller
- It sets out the subject-matter and duration of the processing, the nature and purposes of the processing, the type of personal data and categories of data subjects
- It determines specific tasks and responsibilities of the processor
- It considers the risk to the rights and freedoms of the data subject

DATA TRANSFER OUTSIDE THE EU



PRIVACY NOTICES

Art. 13 and 14

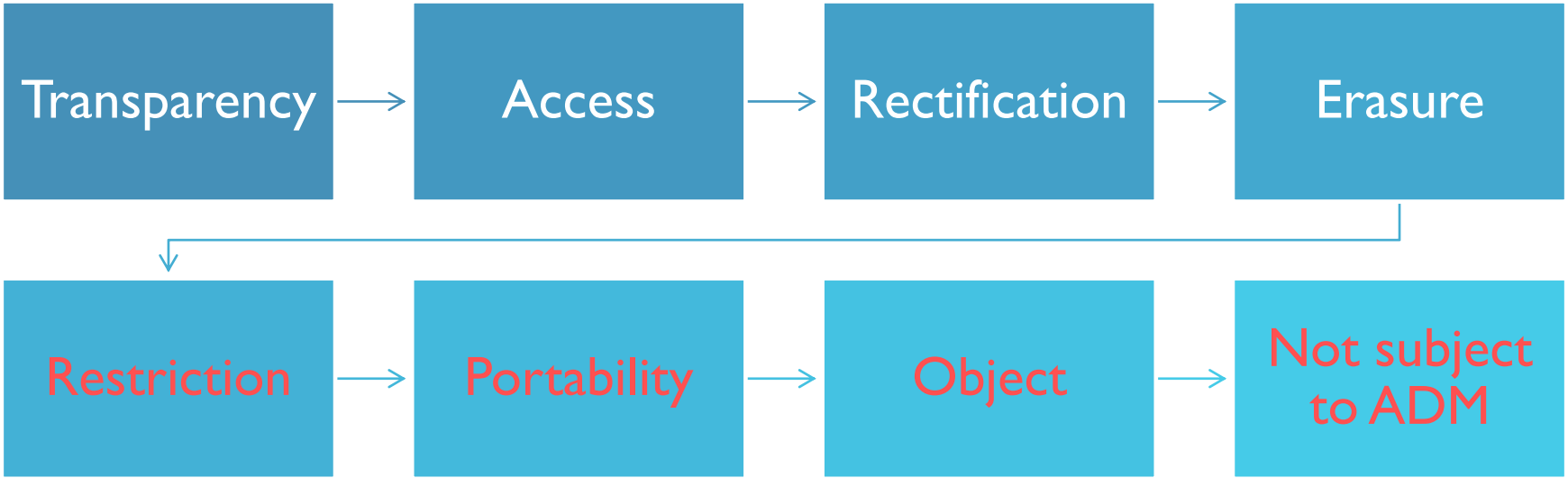
- Clear language
- Simple wording
- Adapting to data subject's level
- Translation into data subject's mother tongue

PRIVACY NOTICES: WHAT SHOULD I WRITE?

Art. 13 and 14

- Name of Controllers, Processors, DPO and source of data
- Categories of data
- Purposes of processing and further processing
- Legal basis
- Retention time
- Transfer of Data and recipients
- ADM and Profiling
- Separate consent boxes
- Rights of the Data Subjects

RIGHTS OF THE DATA SUBJECTS



RIGHTS OF THE DATA SUBJECTS

- Art. 15: access
- Art. 16: rectification
- Art. 17: erasure
- Art. 18: restriction
- Art. 19: notification
- Art. 20: portability
- Art. 21: objection



For **scientific or historical research** purposes or **statistical** purposes, **law** may provide for derogations from **Articles 15, 16, 18 and 21** if they render **impossible** or seriously impair the achievement of the specific purposes



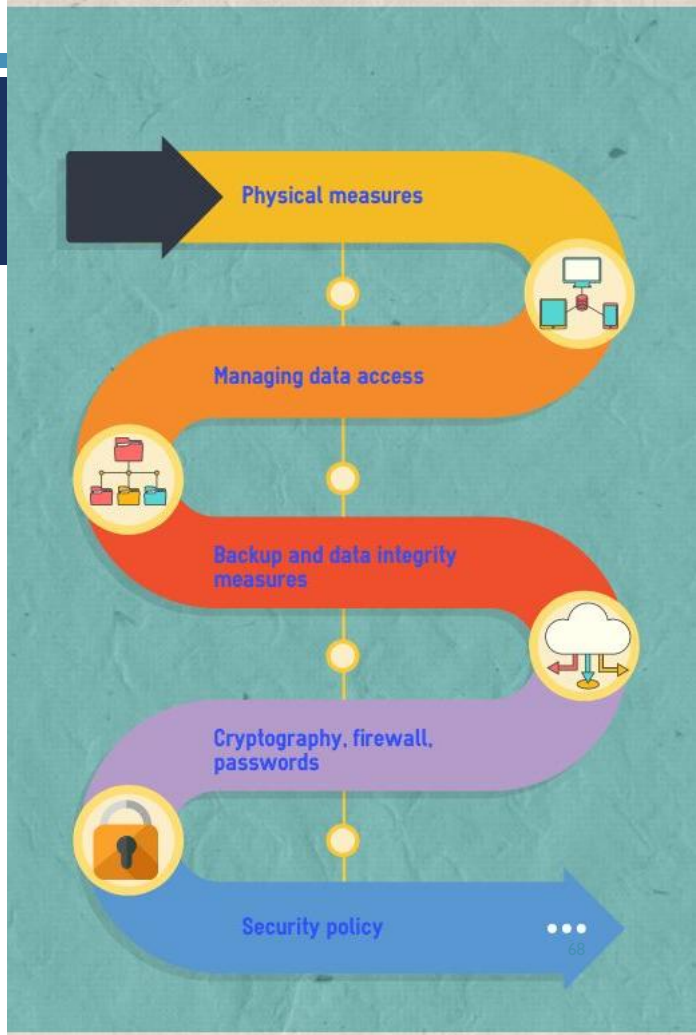
For **archiving** purposes in the **public interest**, **law** may provide for derogations from **Articles 15, 16, 18, 19, 20 and 21** if they render **impossible** or seriously impair the achievement of the specific purposes



Art. 17 shall not apply in some cases when the processing is necessary for **freedom of information, public interest, scientific research**, or for **legal claims**

SECURITY MEASURES

- **Technical and organizational measures**
- **Best practices** of data governance
- Considering the **size** of the organization, the **budget** available, the **type** of data
- Assess the **risks** for data subject
- Privacy **by design** – Privacy **by default** principles
- **Penetration tests**
- Regular **audits**



RISK ASSESSMENT

Likelihood and impact

- Destruction
- Loss
- Unauthorized change
- Unauthorized access
- Unauthorized disclosure

ANONYMIZATION



ANONYMOUS, ANONIMIZED, PSEUDONYMIZED

Anonymous data: there were no identifying data in the first place

Anonymized data: identifying data were removed

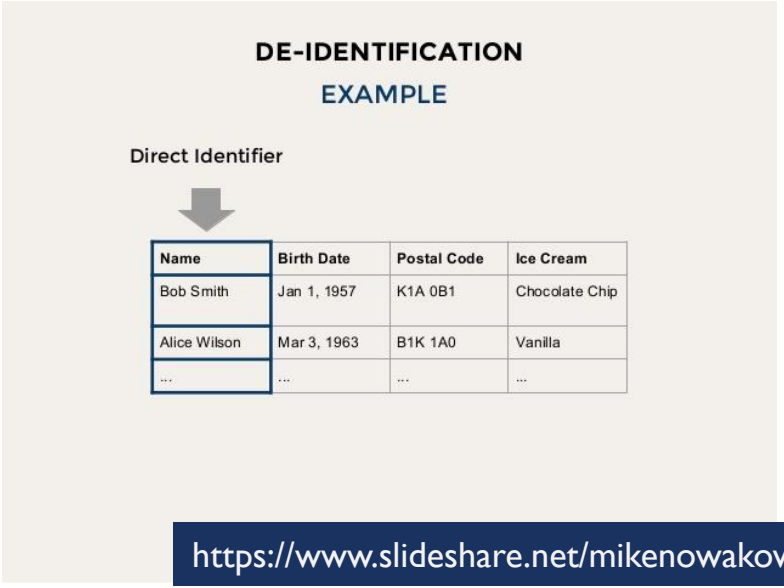
Pseudonymized data: it is still possible to re-identify data using a key

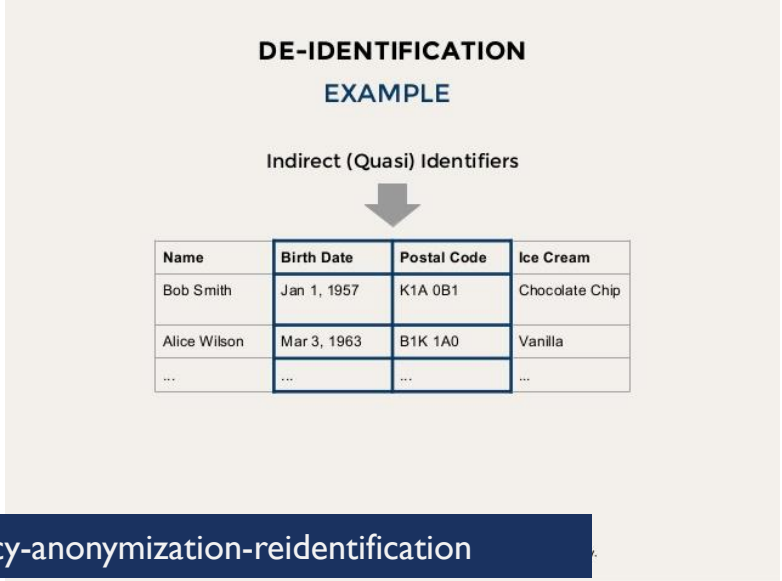
ARE ANONYMIZED DATA TRULY ANONYMIZED?

"In 2008, an anonymised Netflix dataset of film ratings was deanonymised by comparing the ratings with public scores on the IMDb film website in 2014; the home addresses of New York taxi drivers were uncovered from an anonymous data set of individual trips in the city; and an attempt by Australia's health department to offer anonymous medical billing data could be reidentified by cross-referencing "mundane facts" such as the year of birth for older mothers and their children, or for mothers with many children."

Read more at: 'Anonymised' data can never be totally anonymous, says study | Data protection | The Guardian

EXAMPLES





https://www.slideshare.net/mikenowakowski/data-privacy-anonymization-reidentification

K-Anonymity: https://www.researchgate.net/figure/K-anonymized-Data-Set-k3_tbl1_281467551

EXAMPLES

DE-IDENTIFICATION

EXAMPLE

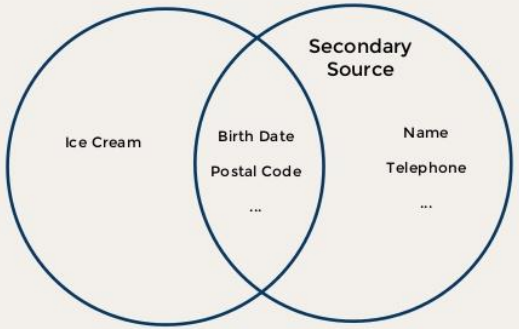
ID	Name	Birth Date	Postal Code	Ice Cream
47562	Bob Smith	Jan 1, 1976	K1A 0B1	Chocolate Chip
17236	Alice Wilson	Mar 3, 1963	B1K 1A0	Vanilla
...	...	...	...	

ID	Ice Cream
47562	Chocolate Chip
17236	Vanilla
...	...

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RE-IDENTIFICATION

LINKAGE



This presentation content is for educational and information purposes only.

ANONYMIZATION TECNIQUES

Evaluation of main techniques

	Is singling out still a risk ?	Is linkability still a risk ?	Is inference still a risk?
Pseudonymisation	Yes	Yes	Yes
Noise addition	Yes	May not	May not
Substitution	Yes	Yes	May not
Aggregation or K-anonymity	No	Yes	Yes
L-diversity	No	Yes	May not
Hashing/ Tokenization	Yes	Yes	May not

No single technique eliminates all risks

No singling out

No linkability

No inference

Source: LawSG-20180329 GDPR Dr Jambert Talk (panopto.eu)

CASE STUDIES

Are those data correctly anonymized?

Assigned ID	Gender	Year of birth (age)	Disease	Hospital	Date of admittance
A001	F	1979	Renal calculi	Eindhoven	19-03-2017
A002	M	1955	Prostatic cancer	Eindhoven	20-03-2017
A003	M	1955	Prostatic cancer	Eindhoven	18-06-2017
A004	M	1980	Renal calculi	Eindhoven	08-09-2017
A005	F	1980	Renal calculi	Eindhoven	01-12-2017
A006	F	1955	Prostatic cancer	Eindhoven	27-12-2017

CASE STUDIES

Are those data correctly anonymized?

Assigned ID	Gender	Year of birth (age)	Disease	Hospital	Date of admittance
A001	F	1979	Renal calculi	Tilburg	19-03-2017
A002	M	1955	Prostatic cancer	Eindhoven	19-03-2017
A003	M	1955	Prostatic cancer	Helmond	19-03-2017
A004	M	1980	Renal calculi	Eindhoven	01-12-2017
A005	F	1980	Renal calculi	Eindhoven	01-12-2017
A006	F	1955	Prostatic cancer	Eindhoven	01-12-2017

These data are not anonymous but **pseudonymized**. The hospital still retains the patients' names, and it is possible to single out specific patients among the others. In this example, the last patient is clearly a trans woman.

4. THE AI ACT

What is considered AI in the new regulation?



a) **Machine learning approaches**, including supervised, unsupervised and reinforcement learning, using a wide variety of methods including deep learning

b) **Logic- and knowledge-based approaches**, including knowledge representation, inductive (logic) programming, knowledge bases, inference and deductive engines,(symbolic) reasoning and expert systems

c) **Statistical approaches**, Bayesian estimation, search and optimization methods

AMENDMENTS TO THE DEFINITION – ARTICLE 3

«'artificial intelligence system' (AI system) means a system that is designed to operate with **elements of autonomy** and that, based on machine and/or human-provided data and inputs, infers how to achieve a given set of objectives using **machine learning and/or logic- and knowledge based approaches**, and produces system-generated outputs such as **content (generative AI systems), predictions, recommendations or decisions**, influencing the environments with which the AI system interacts»

The protection of citizens' rights

The most important innovation of the proposal is the establishment of **four risks categories for AI systems**, in order to protect citizens' fundamental rights. The explanatory memorandum attached to the proposal, in fact, notes that the use of AI with its specific characteristics (e.g. opacity, complexity, dependency on data, autonomous behaviour) can adversely affect a number of fundamental rights enshrined in the EU Charter of Fundamental Rights. The proposal seeks to ensure a high level of protection for those fundamental rights and aims to address various sources of risks through a clearly defined **risk-based approach**.

Classification of AI systems

The risk categories are related to the degree (intensity and scope) of risk for the safety or fundamental rights of citizens and are classified in **three** different groups:

(i) unacceptable risk

(ii) high risk

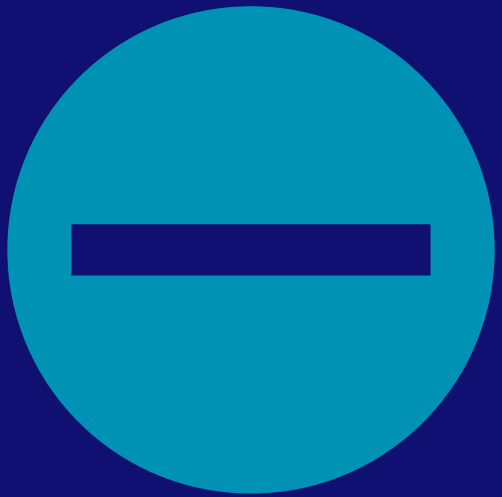
(iii) low or minimal risk (+1 = no risk)

Taking inspiration from the product safety legislation, the classification of risks is based on the intended purpose and modalities for which the AI system is used, not only on their specific function. Depending on the national legal system, the qualification of high risk may have consequences over liability, such as that under art. 2050 of the Italian Civil Code.

Banned technology

Some AI systems are no longer allowed in the EU, with few exceptions, such as those that:

- deploy **subliminal techniques** beyond a person's consciousness in order to materially **distort** a person's **behaviour**
- **exploit** any of the **vulnerabilities** of a specific group of persons due to their age, physical or mental disability
- Are used by the P.A. for the **evaluation** or classification of the **trustworthiness** of natural persons based on their social behaviour or their characteristics
- use **'real-time' remote biometric identification** systems in **publicly accessible spaces** for the purpose of **law enforcement**



Biometric identification systems for law enforcement

- use **'real-time' remote biometric identification** systems in **publicly accessible spaces** for the purpose of **law enforcement**: unacceptable risk

Remote biometric identification systems are defined as an AI systems 'for the purpose of identifying natural persons at a distance through the comparison of a person's biometric data with the biometric data contained in a reference database, and without prior knowledge of the user of the AI system whether the person will be present and can be identified' (Article 3(36)).

The Proposal distinguishes 'real-time' and 'post' remote biometric identification. *Real-time* remote biometric identification systems are defined as AI systems 'whereby the capturing of biometric data, the comparison and the identification all occur without a significant delay'. This definition introduces a mechanism to avoid circumvention, specifying that *real-time remote biometric identification systems* encompass both 'instant identification' and 'limited short delays' (Article 3(37)). *Post* remote biometric identification systems stand for any other type of remote identification system (Article 3(38)).

Real time identification

Only systems that fulfill these requirements are banned:

- only **'real-time' remote biometric identification** (not ex post identification)
- systems used in **publicly accessible spaces** (not private spaces)
- for the purpose of **law enforcement** (not other purposes)

There are also exceptions, including the search of victims, the prevention of threats to the life or safety of individuals, or the identification of a perpetrator or suspect of a criminal offense (punished with a minimum of 3 years of conviction), subject to a judicial review most of the time.

Biometric data

Data relating to the physical, physiological or behavioral characteristics of an individual which allow their unique identification, such as facial images or dactyloscopic data.

You need to be particularly careful when you handle this type of data

EDPB guidelines

Guidelines 3/2019 on videosurveillance

The intensive use of video devices has an impact on citizen's behaviour. Significant implementation of such tools in many spheres of the individuals' life will put an additional pressure on the individual to prevent the detection of what might be perceived as anomalies. De facto, these technologies may limit the possibilities of anonymous movement and anonymous use of services and generally limit the possibility of remaining unnoticed. Data protection implications are massive.

Risks of biometrical profiling

Privacy issues: intrusive facial recognition/ profiling of behaviours in public and private spaces. Video surveillance systems usually collect massive amounts of personal data which may reveal data of a highly personal nature and even special categories of data. Indeed, apparently non-significant data originally collected through video can be used to infer other information to achieve a different purpose (e.g. to map an individual's habits). Biometric identification practices necessarily affect individuals who have no reason to be observed.

In addition to privacy issues, there are also risks related to possible malfunctions of these devices and the biases they may induce. Researchers report that software used for facial identification, recognition, or analysis performs differently based on the age, gender, and ethnicity of the person it's identifying.

Algorithms would perform based on different demographics, thus, bias in facial recognition threatens to reinforce the prejudices of society. That is why, data controllers must also ensure that biometric data processing deriving from video surveillance be subject to regular assessment of its relevance and sufficiency of guarantees provided

RECITAL 33

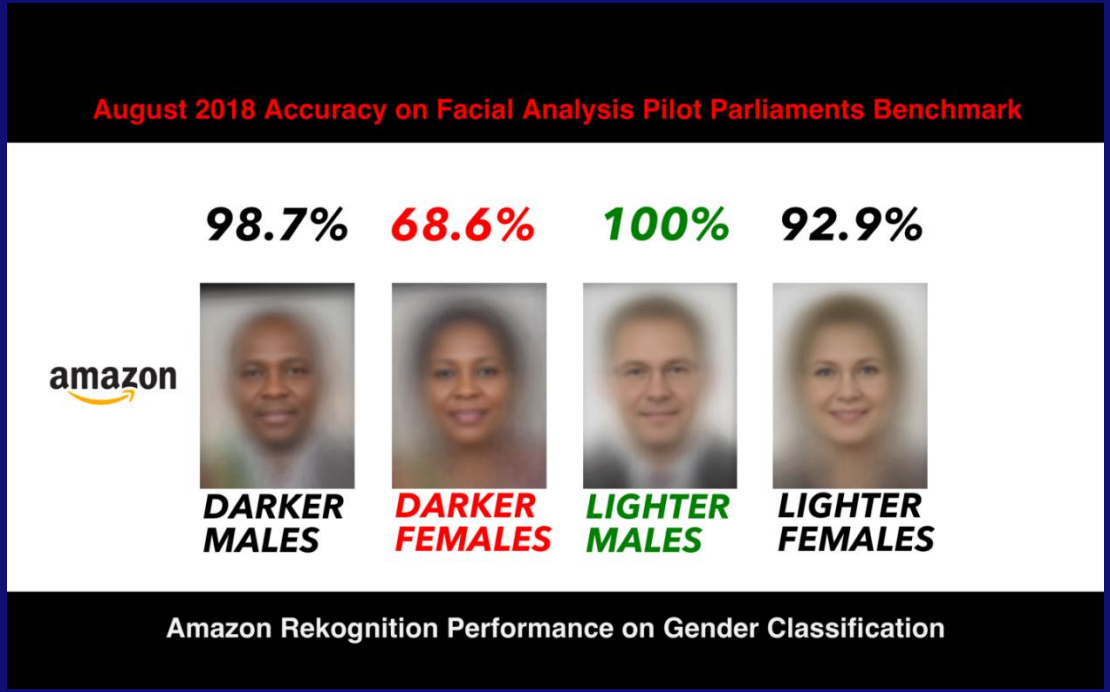
«Technical inaccuracies of AI systems intended for the remote biometric identification of natural persons can lead to **biased results** and entail discriminatory effects. This is particularly relevant when it comes to age, ethnicity, race, sex or disabilities. Therefore, 'real-time' and 'post' remote biometric identification systems should be classified as high risk. In view of the risks that they pose, both types of remote biometric identification systems should be subject to specific requirements on logging capabilities and human oversight.»

What is bias?

The term bias is used by Friedman and Nissenbaum, who say it "refers to computer systems that **systematically and unfairly discriminates against certain individuals or groups of individuals in favor of others**". By extension pre-existing societal bias is "**bias that originates from society at large, such as from organizations (e.g., industry), institutions (e.g., legal systems), or culture at large**. (E.g., gender biases present in the larger society that lead to the development of educational software that overall appeals more to boys than girls)". Pre-existing bias can be implicit and subconscious, rather than malicious in intent.

Source: "Does Technology Have Race?'', Hankerson et al, 2016

Examples

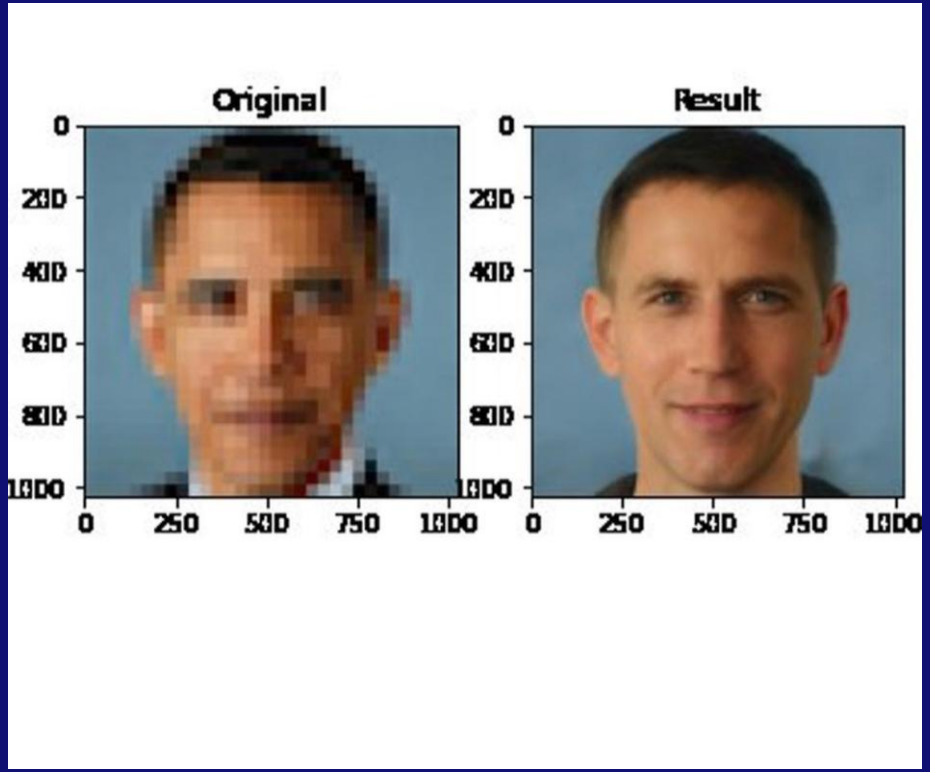

August 2018 Accuracy on Facial Analysis Pilot Parliaments Benchmark
98.7%
68.6%
100%
92.9%
amazon
DARKER MALES
DARKER FEMALES
LIGHTER MALES
LIGHTER FEMALES
Amazon Rekognition Performance on Gender Classification

Examples

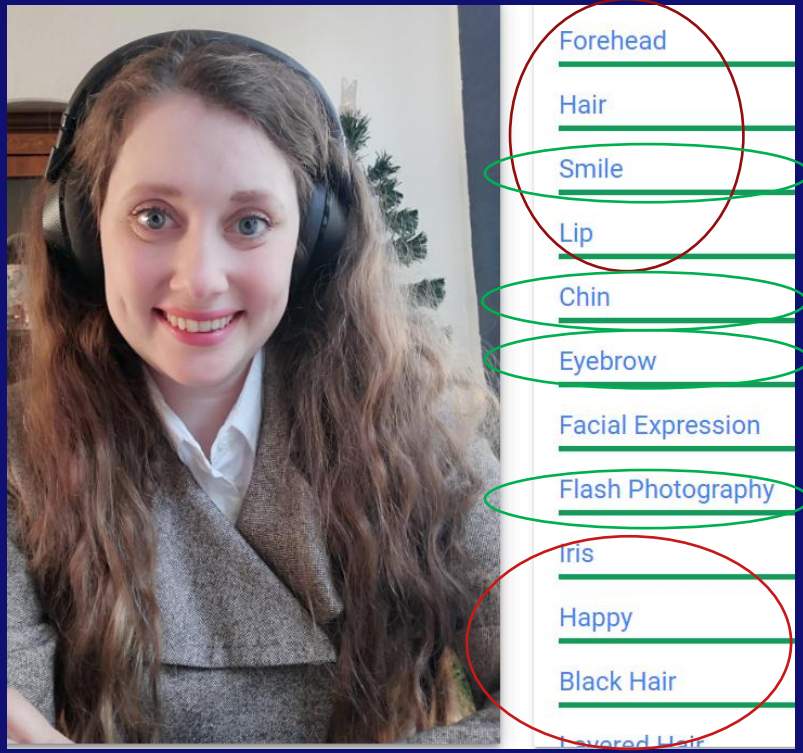


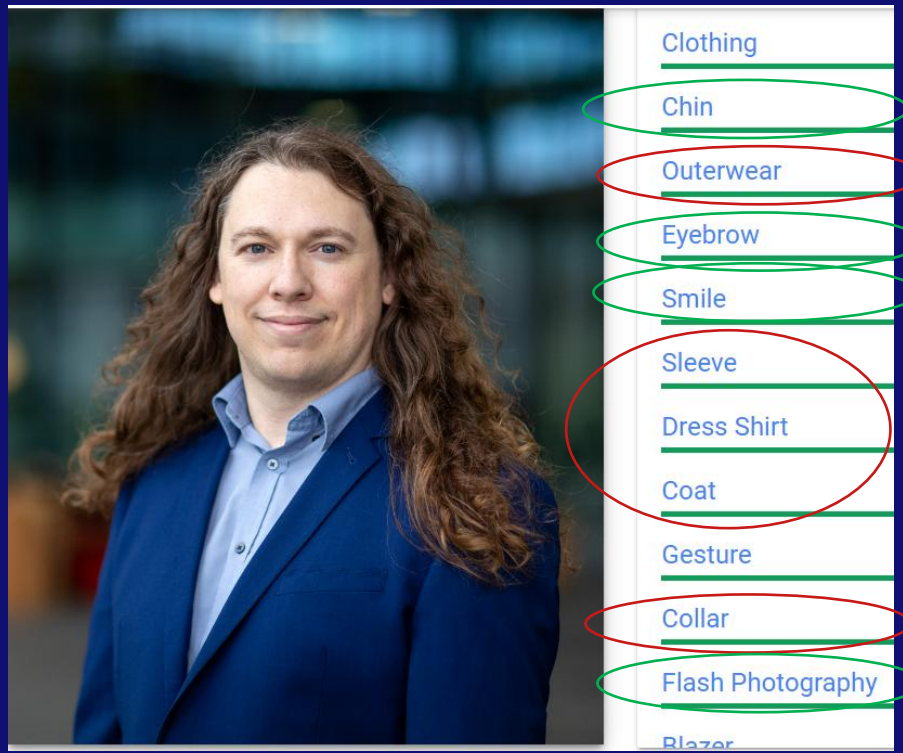
Google and its gorillas

Examples

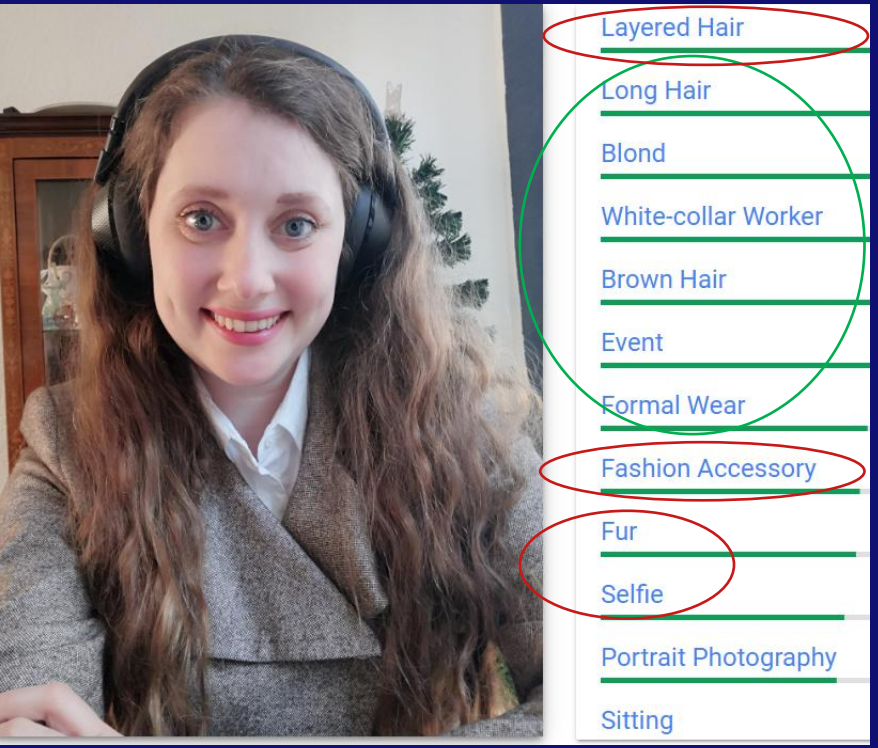


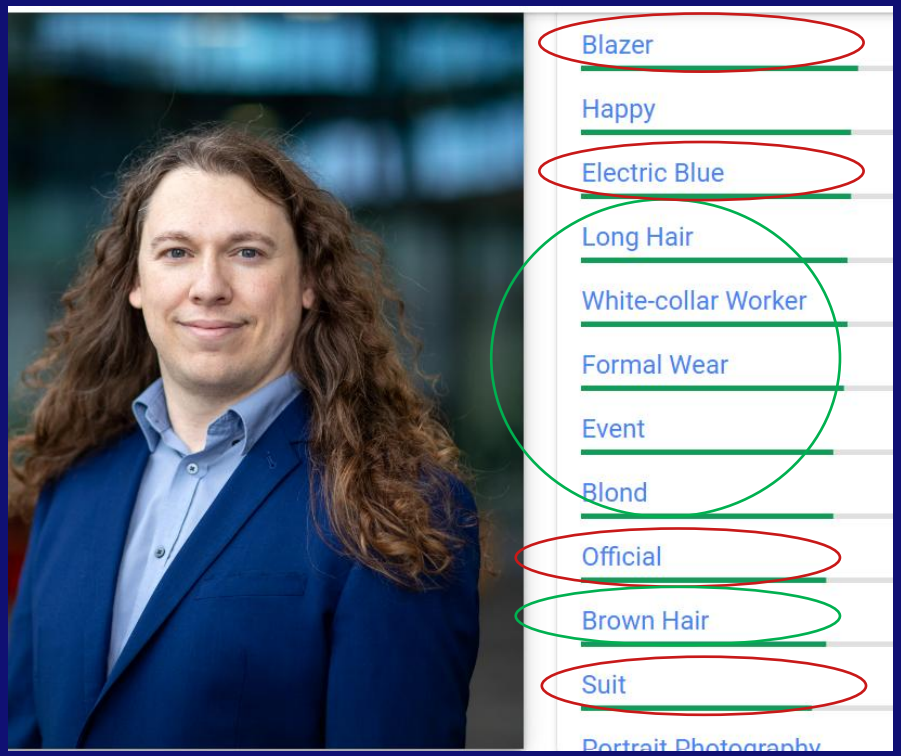
Examples


Forehead
Hair
Smile
Lip
Chin
Eyebrow
Facial Expression
Flash Photography
Iris
Happy
Black Hair


Clothing
Chin
Outerwear
Eyebrow
Smile
Sleeve
Dress Shirt
Coat
Gesture
Collar
Flash Photography

Examples


Layered Hair
Long Hair
Blond
White-collar Worker
Brown Hair
Event
Formal Wear
Fashion Accessory
Fur
Selfie
Portrait Photography
Sitting


Blazer
Happy
Electric Blue
Long Hair
White-collar Worker
Formal Wear
Event
Blond
Official
Brown Hair
Suit

Examples



Portrait Photography
Sitting
Hair Coloring
Uniform
Portrait
Laugh
Jewellery
Pattern
Photo Shoot
Feathered Hair
Official



Suit
Portrait Photography
Portrait
Uniform
Job
Button
Beard
Spokesperson
Businessperson
Business
Employment

WHY DO WE CARE ABOUT BIAS?

- 1) Because it can lead to discrimination towards individuals or groups
- 2) Because it can lead to inaccurate results
- 3) Because it can cause errors and damage to people
- 4) Because it can cause negative effects on society

TYPES OF BIASES AND ERRORS IN DATA SETS & MODELS

- Selection bias
- Overfitting and underfitting the data
- Reporting Biases
- Overgeneralization Bias
- Group Attribution Biases
- Implicit Biases
- Variance
- Automation bias

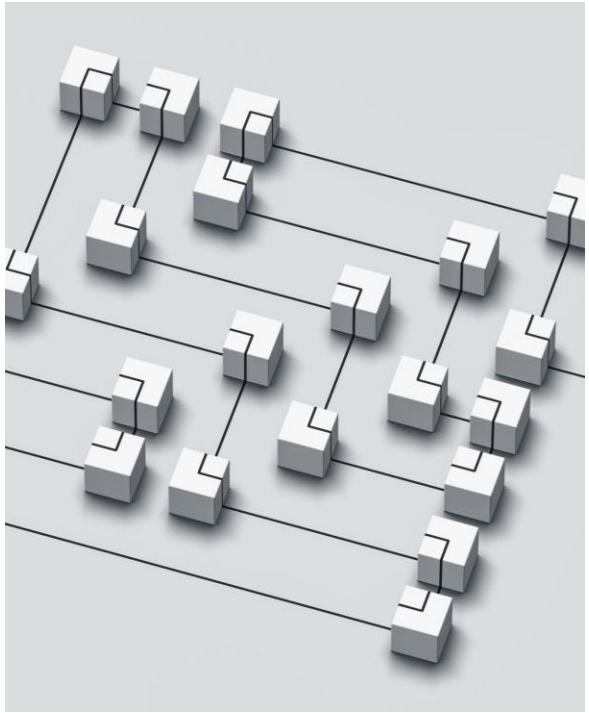
SELECTION BIAS

When data are selected in a way that is not representative of the distribution of data in real life, selection bias occurs. This occurs as a result of improper randomization during data collection.

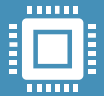
When randomization is improperly achieved during data collection, **sampling bias** occurs.

When data are not chosen in a representative way, **convergence bias** occurs. For instance, your dataset does not accurately reflect the group of people who did not buy your goods if you just survey the half of your consumers who actually bought your product.

Participation bias is when there are gaps in involvement during the data collection process, making the data unrepresentative.



OVER- AND UNDER- FITTING



In machine learning, overfitting occurs when a model is trained with so much data that it begins to learn from the noise and inaccurate data entries in the data set. Machine learning models have trouble predicting new data based on the training data because what they learnt from the noise cannot be applied to new data.



When a machine learning model fails to capture the underlying trend of the data, underfitting occurs. In this case, it indicates that the model or algorithm is not fitting the data well enough.

REPORTING BIAS

The inclusion of only a portion of the results in an analysis, which often only includes a small portion of the evidence, is referred to as reporting bias, which can appear in many different ways.

Citation bias happens when your analysis is dependent on research that were included in other studies' citations.

When you disregard news sources that are not in your native tongue, you are exhibiting language bias.

When studies are given higher weight because they have been published more than once, this is known as duplicate publication bias.

Publication bias is the tendency for research with favorable results to be published more frequently than those with unfavorable outcomes or insufficient evidence.

Bias in the reporting of outcomes happens when some results are selectively reported. In a quarterly report, for instance, you only disclose when the company reports positive results.

Time lag bias is a problem when studies take a long time to be published.

OVERGENERALIZATION BIAS

Overgeneralization occurs when someone extrapolates from one occurrence to all subsequent events, thinking that what you observe in one dataset will also be observed in another.

It happens when you think that, regardless of the dataset's size, what you observe in your dataset is what you would also see if you looked in any other dataset intended to evaluate the same information.

GROUP ATTRIBUTION BIAS

Group attribution biases are the tendency for people to think that a group's beliefs always dictate an individual's traits or that a group's decisions are impacted by the emotions of all of its members.



.

When you favor those in a group to which you personally belong or have similar interests, you are exhibiting in-group bias.

When you stereotype certain members of a group to which you do not directly belong, this is known as out-group bias.

IMPLICIT BIAS

When assumptions are drawn from personal experiences that do not necessarily apply to more generic situations, implicit bias results. Without realizing it, people frequently act with bias and stereotypes in mind. Even when we are not aware of it, implicit bias takes the form of opinions and prejudices we have about other people. We could seek for information that confirms our assumptions and ideas while ignoring information that contradicts them.

We might associate, for instance, the white color to peace or purity. White, however, in Chinese and Japanese culture is associated to funerals.

The propensity to look for information in a way that supports or confirms one's preexisting opinions or experiences is known as confirmation bias or experimenter's bias.

VARIANCE

The variability in a model's prediction is known as variance. High variance models, in contrast to biased models, focus excessively on training data and fail to generalize to new data. On training data, this produces positive outcomes, but on test data, it produces high error rates. Variance can be thought of as the model's sensitivity to changes in the data. From noise, our model might learn. This will lead our model to value unimportant features highly. Low bias and low variance machine learning models are impossible to produce. However, you can try to make a tradeoff between the bias and the variance.

AUTOMATION BIAS

Automation bias is the propensity for people to favor outcomes or recommendations produced by automated systems and to discount contradictory data from non-automated systems, even when it is accurate. It happens when you act upon this AI-based recommendation without first checking to see if the data was accurate.

For instance, a data scientist relies on the analytics platform's advice to edit or eliminate specific data despite the platform's warnings, which lowers the data's quality.



BIAS IS AN IMPORTANT CONCEPT IN THE AI ACT – RECITAL 44

"High data quality is essential for the performance of many AI systems, especially when techniques involving the training of models are used, with a view to ensure that the high-risk AI system performs as intended and safely and it does not become the source of discrimination prohibited by Union law. High quality training, validation and testing data sets require the implementation of appropriate data governance and management practices. Training, validation and testing data sets should be sufficiently relevant, representative and have the appropriate statistical properties, including as regards the persons or groups of persons on which the high-risk AI system is intended to be used. These datasets should also be as free of errors and complete as possible in view of the intended purpose of the AI system, taking into account, in a proportionate manner, technical feasibility and state of the art, the availability of data and the implementation of appropriate risk management measures so that possible shortcomings of the datasets are duly addressed. The requirement for the datasets to be complete and free of errors should not affect the use of privacy-preserving techniques in the context of the development and testing of AI systems. Training, validation and testing data sets should take into account, to the extent required by their intended purpose, the features, characteristics or elements that are particular to the specific geographical, behavioural or functional setting or context within which the AI system is intended to be used. In order to protect the right of others from the discrimination that might result from the bias in AI systems, the providers should be able to process also special categories of personal data, as a matter of substantial public interest within the meaning of Article 9(2)(g) of Regulation (EU) 2016/679 and Article 10(2)g) of Regulation (EU) 2018/1725, in order to ensure the bias monitoring, detection and correction in relation to high-risk AI systems."

The concept of ‘High Risk’

Some AI system are considered **high risk** and have to meet certain requirements before being released to the market or to the public. Some examples includes systems used in:

- Machinery (including robots)
- Toys
- Medical devices
- In Vitro Diagnostic
- Vehicles
- Aircrafts
- Rails



AI Regulation must be coordinated with MDR, GDPR, and other laws (e.g., civil liability rules)

Other High Risk Systems

Biometric identification and categorisation of natural persons	Management and operation of critical infrastructure	Education and vocational training	Employment, workers management and access to self-employment
Access to and enjoyment of essential private services and public services and benefits	Law enforcement	Migration, asylum and border control management	Administration of Justice

Mandatory requirements for High Risk Systems

1. **Risk management system**:

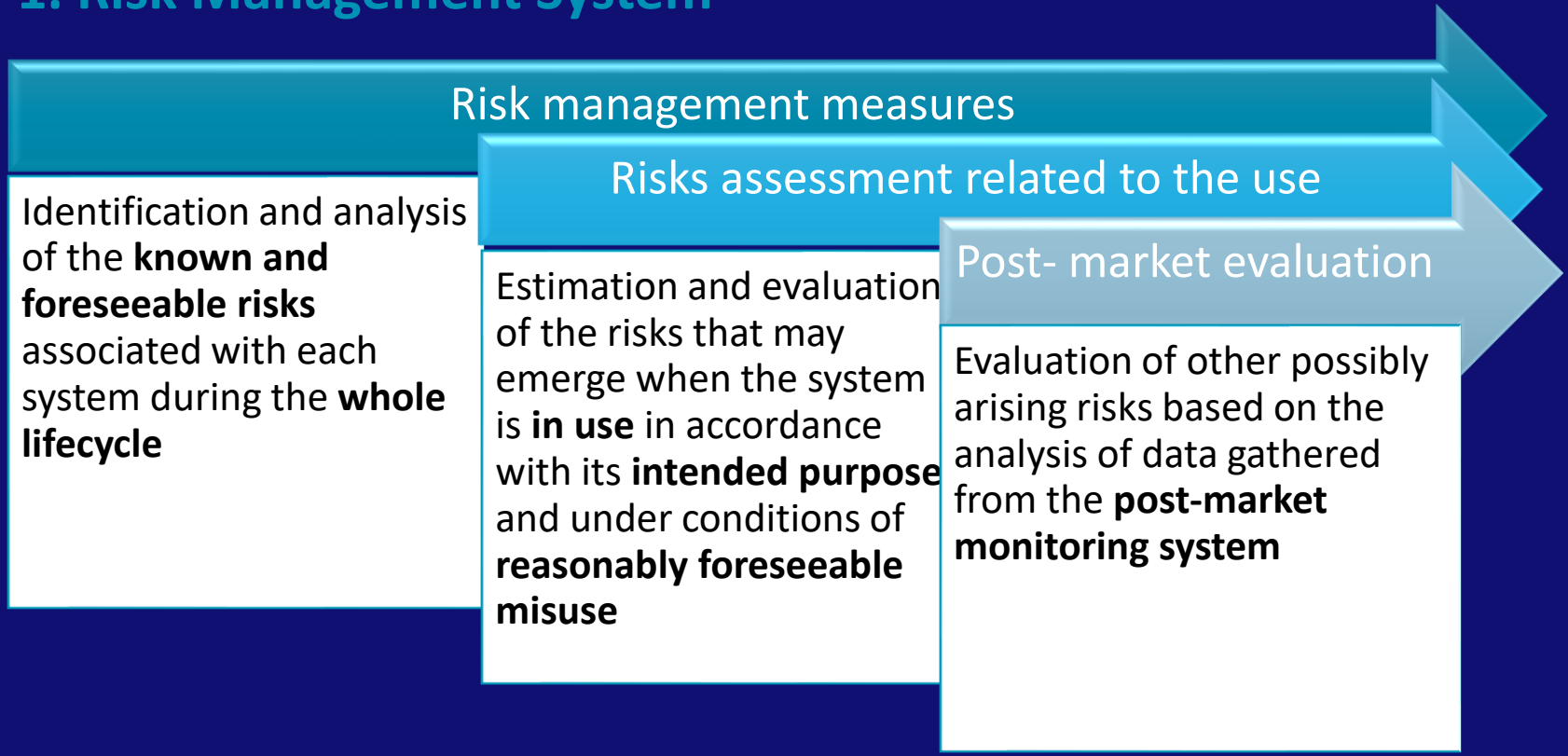
a) identification and analysis of the **known and foreseeable risks** associated with each system

b) estimation and evaluation of the risks that may emerge when the system is used in accordance with its **intended purpose** and under conditions of **reasonably foreseeable misuse**

c) evaluation of other possibly arising risks based on the analysis of data gathered from the **post-market monitoring system**

d) adoption of **suitable risk management measures**

1. Risk Management System



Mandatory requirements for High Risk Systems

2. **Data and data governance**:

Training, validation and testing data sets concerning:

(a) the relevant **design choices**
(b) **data collection**
(c) relevant **data preparation processing operations**, such as annotation, labelling, cleaning, enrichment and aggregation
(d) the formulation of **relevant assumptions**, notably with respect to the information that the data are supposed to measure and represent
(e) a prior assessment of the **availability, quantity and suitability of the data sets** that are needed
(f) examination in view of possible **biases**
(g) the identification of any possible **data gaps or shortcomings**, and how those gaps and shortcomings can be addressed

2. Data Management

Design
- Design choices/ model choice
- Privacy by design and by default
- Interpretable/Explainable AI

Collection
- GDPR
- Copyright
- Re-use assessment

Preparing
- Annotation, labelling, cleaning, enrichment and aggregation of data
- Data set validation

Assumptions
- Information that the data are supposed to measure and represent
- Features/variables

Assessment
- Availability, quantity and suitability of the data sets
- Testing if data are sufficient/ accurate

Biases
- Assessment for biases/discrimination
- Testing to check if biases are present

Gaps
- Identification of gaps and shortcomings
- Measures to fill and correct them
- Testing if measures are effective

ARTICLE 10

«To the extent that it is strictly necessary for the purposes of **ensuring bias monitoring, detection and correction** in relation to the high-risk AI systems, the providers of such systems may process **special categories of personal data** referred to in Article 9(1) of Regulation (EU) 2016/679 [GDPR], Article 10 of Directive (EU) 2016/680 [LED Directive] and Article 10(1) of Regulation (EU) 2018/1725 [Data protection for public bodies], subject to **appropriate safeguards** for the fundamental rights and freedoms of natural persons, including technical limitations on the re-use and use of state-of-the-art security and privacy-preserving measures, such as pseudonymisation, or encryption where anonymisation may significantly affect the purpose pursued.»

Mandatory requirements for High Risk Systems

3. **Technical documentation**:

1. Drawn **before releasing** the system

2. **Demonstrate compliance** with the requirements set out in the Regulation and provide national competent authorities and notified bodies with all the necessary information to assess the compliance with those requirements

3. Enabling the **automatic recording of events** (‘logs’) while the systems is operating

Mandatory requirements for High Risk Systems

5. **Transparency** and provision of **information** to users:

(a) the **identity and the contact** details of the provider and, where applicable, of its authorised representative;

(b) the **characteristics, capabilities and limitations** of performance of the system

(c) the **changes** to the system and its performance which have been pre-determined by the provider at the moment of the initial conformity assessment, if any;

(d) the **human oversight measures**, including the technical measures put in place to facilitate the interpretation of the outputs of systems by the users;

(e) the **expected lifetime** of the system and any **necessary maintenance** and care measures to ensure the proper functioning of that AI system, including as regards **software updates**

Mandatory requirements for High Risk Systems

6. **Human oversight**

Through measures:

(a) identified and **built into the system**, when technically feasible, by the provider **before** it is placed on the market or put into service, and/or

(b) identified by the provider **before** placing the system on the market or putting it into service and that are appropriate **to be implemented by the user**

Mandatory requirements for High Risk Systems

7. **Accuracy, robustness and cybersecurity**:

Systems shall be **resilient as regards errors, faults or inconsistencies** that may occur within the system or the environment in which the system operates

The robustness of systems may be achieved through **technical redundancy solutions**, which may include backup or fail-safe plans

Systems shall be **resilient as regards attempts by unauthorised third parties** to alter their use or performance by exploiting the system vulnerabilities

The technical solutions to address AI specific vulnerabilities shall include, where appropriate, measures to **prevent and control for attacks** trying to manipulate the training dataset (‘data poisoning’), inputs designed to cause the model to make a mistake (‘adversarial examples’), or model flaws

RECITAL 50

"The technical robustness is a key requirement for high-risk AI systems. They should be **resilient** in relation to **harmful or otherwise undesirable behaviour** that may result from limitations within the systems or the environment in which the systems operate (e.g. errors, faults, inconsistencies, unexpected situations). High-risk AI systems should therefore be designed and developed with appropriate technical solutions to prevent or minimize that harmful or otherwise undesirable behaviour, such as for instance mechanisms enabling the system to safely interrupt its operation (fail-safe plans) in the presence of certain anomalies or when operation takes place outside certain predetermined boundaries. Failure to protect against these risks could lead to safety impacts or **negatively affect the fundamental rights**, for example due to erroneous decisions or wrong or **biased outputs** generated by the AI system"

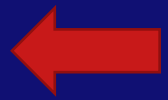
ARTICLE 15

«High-risk AI systems that continue to learn after being placed on the market or put into service shall be developed in such a way to **eliminate or reduce as far as possible** the risk of possibly **biased outputs** influencing input for future operations ('feedback loops') are duly addressed with appropriate mitigation measures.»

Mandatory requirements for High Risk Systems

8. Quality management system:

a) a strategy for **regulatory compliance**
b) techniques, procedures and systematic actions for the **design, design control and design verification of the system**
c) techniques, procedures and systematic actions for the **development, quality control and quality assurance** of the system
d) examination, test and validation procedures
e) technical specifications. including standards. to be applied
f) systems and procedures for **data management**, including data collection, data analysis, data labelling, data storage, data filtration, data mining, data aggregation, data retention and any other operation regarding the data that is performed **before** - and for the purposes of - the placing on the market or putting into service of the systems (see n. 2)



Mandatory requirements for High Risk Systems

8. Quality management system:

g) the setting-up, implementation and maintenance of a **post-market monitoring system**
h) procedures related to the **reporting of serious incidents and malfunctioning**
i) the handling of **communication with competent authorities**, including sectoral ones, **providing or supporting the access to data**, notified bodies, other operators, customers or other interested parties
j) systems and procedures for **record keeping** of all relevant documentation and information
k) resource management, including security of supply related measures
l) an **accountability framework** setting out the responsibilities of the management and other staff with regard to all aspects listed in this paragraph

9. **Providers shall draw up technical documentation required by the Regulation**

Transparency obligations for certain AI systems



Providers shall ensure that AI systems **intended to interact with natural persons** are designed and developed in such a way that natural persons are **informed that they are interacting with an AI system**, unless this is obvious from the circumstances and the context of the use



Users of an AI system that **generates or manipulates image, audio or video content** that appreciably **resembles existing persons, objects, places or other entities or events** and would falsely appear to a person to be authentic or truthful (‘**deep fake**’), shall **disclose** that the content has been artificially generated or manipulated

2. BASIC PRINCIPLES OF AI AUDITING

What is AI auditing?

AI Auditing is the **review** of the whole AI lifecycle, including the post-market phase, with the aim of identifying and addressing **risks** for citizens and the society. It is carried out through applying legal, ethical and technical principles in practice.

«*An ethics-based auditing can improve the quality of decision making, increase user satisfaction, unlock growth potential, enable law-making, and relieve human suffering*»

- Mokander & Floridi

How to perform AI auditing?

An AI Audit is carried out through the practical implementation of several principles:

- **Ethics**: checking if the consequences on individuals and society have been taken into account
- **Human Rights**: checking if fundamental rights of citizens are respected
- **Law**: checking if legal compliance has been performed (privacy, cybersecurity, national laws, etc.)
- **Technical good practices**: checking if the system is sound, safe, secure, correct, and biasless (e.g.: avoid sample bias)

Guidance to perform AI auditing

In the literature, several authors have proposed a method to perform an AI audit, but few have shown how to do it in practice.

Most papers focuses on the technical aspects, such as debiasing techniques and data set balancing.

Floridi et al. suggest a guide to implement the new AI Act requirements in its paper «CapAI»

3. BASIC PRINCIPLES OF AI ETHICS

What is AI Ethics?

AI Ethics is a branch of Philosophy; it studies the ethical principles applicable to AI systems and addresses the implications, risks and problems derived from:

1) humans building and employing those systems, and

1) the behavior of autonomous systems

Most famous ethics guidelines

Universal Guidelines for AI (2018)

OECD AI Principles (2019)

G20 AI Guidelines (2019)

UNESCO Recommendation on AI Ethics (2021)

Ethical principles

Table 2 – Ethical principles identified in existing AI guidelines

Ethical principle	Number of documents	Included codes
Transparency	73/84	Transparency, explainability, explicability, understandability, interpretability, communication, disclosure, showing
Justice & fairness	68/84	Justice, fairness, consistency, inclusion, equality, equity, (non-)bias, (non-)discrimination, diversity, plurality, accessibility, reversibility, remedy, redress, challenge, access and distribution
Non-maleficence	60/84	Non-maleficence, security, safety, harm, protection, precaution, prevention, integrity (bodily or mental), non-subversion
Responsibility	60/84	Responsibility, accountability, liability, acting with integrity
Privacy	47/84	Privacy, personal or private information
Beneficence	41/84	Benefits, beneficence, well-being, peace, social good, common good
Freedom & autonomy	34/84	Freedom, autonomy, consent, choice, self-determination, liberty, empowerment
Trust	28/84	Trust
Sustainability	14/84	Sustainability, environment (nature), energy, resources (energy)
Dignity	13/84	Dignity
Solidarity	6/84	Solidarity, social security, cohesion

Jobin, Anna, Marcello Ienca, and Effy Vayena. "The global landscape of AI ethics guidelines." *Nature Machine Intelligence* 1.9 (2019): 389-399.

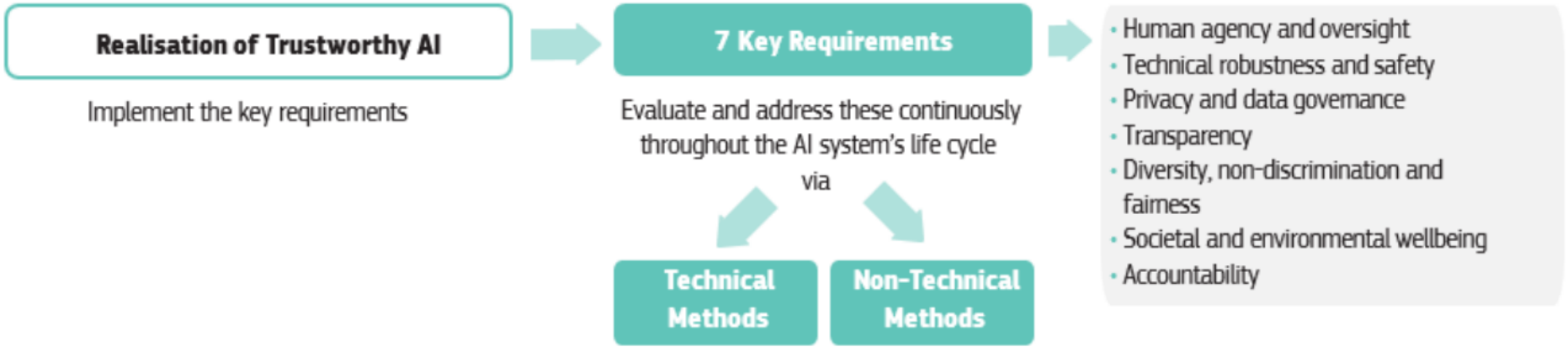
Trustworthy AI

AI HLEG recognizes 3 keys elements of Trusworthy AI:

1. it should be **lawful**, complying with all applicable laws and regulations;
2. it should be **ethical**, ensuring adherence to ethical principles and values; and
3. it should be **robust**, both from a technical and social perspective, since, even with good intentions, AI systems can cause unintentional harm.

The ALTAI List

AI HLEG has drawn a list of 7 elements:





Open Ethics Canvas

This instrument is used by several companies in order to perform an ethical assessment of their AI systems

Green: non-technical ethics
Grey: technical ethical assessment
Yellow: accountability and human oversight
Red: legal assessment and societal liability

TU/e

4. APPLYING ETHICAL PRINCIPLES TO DATA SETS: A PRACTICAL EXAMPLE

Preliminary remarks

An AI system is only as good as the data it relies on.

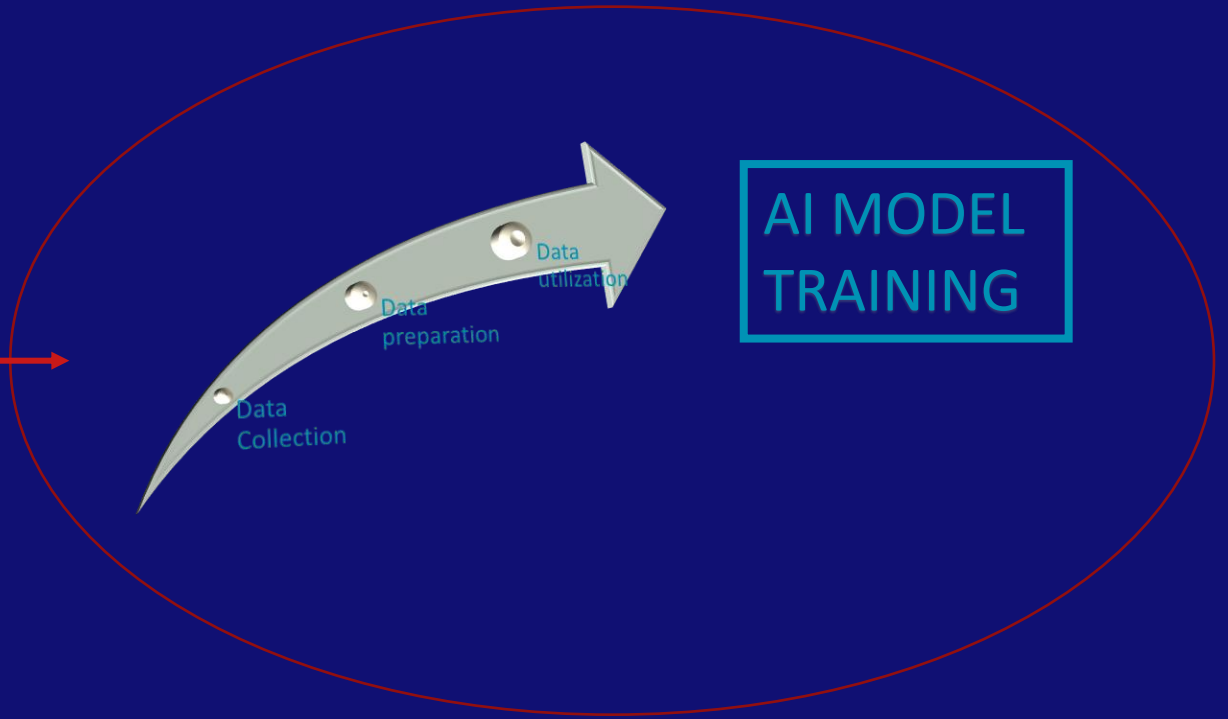
Garbage in = garbage out

Data sets can be unbalanced, biased, incomplete, too small, inadequate for the intended use, not clean and not validated.

It is extremely important to prepare data **before** feeding it to the AI model.

Applying ethical and legal principles in practice

Non-discrimination
Fairness
Transparency
Privacy
Copyright
Accountability
Risk management
Quality assessment
Data Governance



How to apply those principles in practice?





Developing AI models for hearing aids

We are working on a project to develop an AI model to help people with **hearing impairments**

The goal is to **help them hear better and understand what it is said in a conversation**, so the model needs to be able to recognize all kind of voices in all settings

People with hearing impairments have **the right not to be excluded** in a conversation

People with speech impairments, with a different accent or pronunciation, with a low/high tone of voice, or that are slow in talking, have **the right to have their voice recognized**

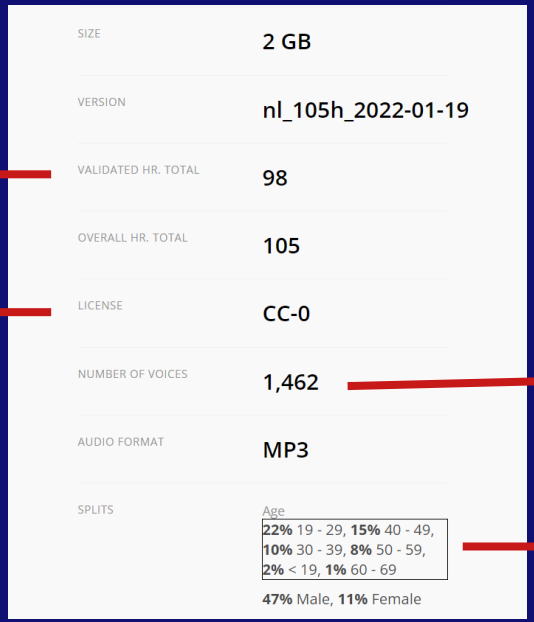
What could go wrong?

If the data set is **biased** and different categories of people are not represented, this could lead to **discrimination** towards both people with hearing impairments and **underrepresented groups**, as the model will **not be able to recognize** all voices






Mozilla Common Voice Data Set



Validation info

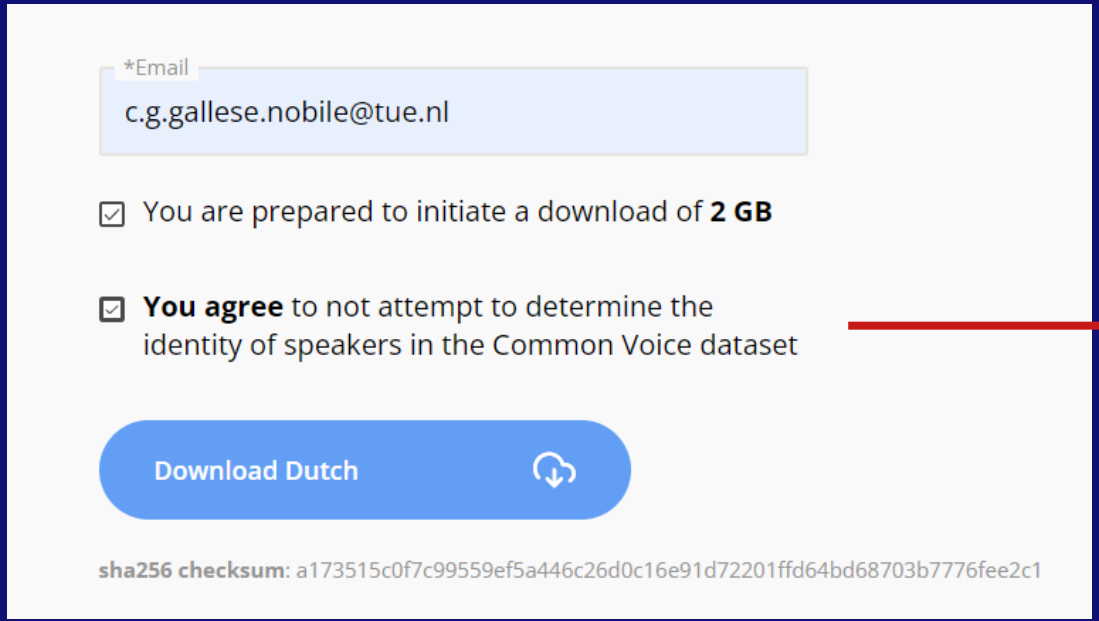
Type of license

Number of examples

Age and gender composition

Intended use?
Type of data?
Model choice?

Terms & Conditions



Privacy

Looking into the data set metadata

	A	B	C	D	E	F	G	H	I	J
1	client_id	path	sentence	up_votes	down_votes	age	gender	accents	ocale	segment
2	0d709133bf209da7f0164653b6e5f9a	common_voice_nl_17699535.mp3	De Aboriginals zijn de oorspronkelijke bewoners van Australi	2	0	fifties	male	Nederland	nl	
3	0fca93407be6d482019f2463e60fbaf	common_voice_nl_17694848.mp3	Mijn toetsenbord zit vol stof.	2	1				nl	
4	175d4117110538cc68a8a0157a7f0a	common_voice_nl_18441136.mp3	Ze had de bank beschadigd met haar skateboard.	2	0				nl	
5	29ecd9ca5970fd1bc1297bb4a9a5f3a	common_voice_nl_30205425.mp3	Deze top heeft niet in alle opzichten aan de verwachtingen vo	2	0				nl	
6	3c448c76f72ee15009da6c9e474dc9f	common_voice_nl_19966492.mp3	Waar laat jij je onderhoud doen?	2	0	thirties	male		nl	
7	452cfa9d7e49e72eac2bb7ff3d5f79c	common_voice_nl_17700547.mp3	Na het lezen van vele beoordelingen had ze eindelijk haar oog	2	1	fifties	male	Nederland	nl	
8	4ac8a9187cfd11da401d6829199c18	common_voice_nl_21571581.mp3	De tampons zijn op.	2	0	twenties	male	Belgisch N	nl	
9	4d6898ec762a9590a973dc27579b93	common_voice_nl_25310936.mp3	Proberen wij kooldioxide-emissies te beperken of proberen wi	2	0				nl	
10	581251914ca5d3eeb5de368076aecc	common_voice_nl_21431804.mp3	Marijke kent Olivier nu al meer dan twee jaar.	2	0	twenties	female	Nederland	nl	
11	5f1e2fabc5a5c63d880ce655697814a	common_voice_nl_28458010.mp3	Mijn partij vindt dat onverkwikkelijk.	2	0				nl	
12	723a65c399699beeb3c730e141775f	common_voice_nl_19031796.mp3	Het voeren van brood aan eenden is eigenlijk ongezond voor d	2	0				nl	
13	77fc542b71acf6510c80e9a681c128f	common_voice_nl_30125083.mp3	Juni dit jaar is veel te laat.	2	0	seventies	male		nl	
14	79d570bd61fb0fcf4d363d0a1e044cf	common_voice_nl_17695742.mp3	Parket moet je stofzuigen, tegels moet je dweilen.	2	1				nl	
15	7aa3af7141d503c7c449404c3f7939	common_voice_nl_19691757.mp3	In onze buurt kent iedereen elkaar.	2	1	teens	male	Nederland	nl	
16	8d9fece2b05d9fe37e9c66e45767cae	common_voice_nl_22759350.mp3	Naar mijn oordeel valt er namelijk niet meer zoveel te verbete	2	0				nl	
17	9062e9bc9c6ee752e6f836d4296ee3	common_voice_nl_30057990.mp3	Daarom wordt voorgesteld dit soort pogingen te verbieden en	2	0				nl	
18	a01a8a30db997795dc07295a000bbc	common_voice_nl_17706390.mp3	Heb je de tafel gedekt?	2	1				nl	
19	a64ddf48b5b3b6700b146de59126a0	common_voice_nl_17696892.mp3	Ine lag in de strandstoel.	2	1				nl	
20	a9f1622fcaedf8cc1fba8d13c58564b	common_voice_nl_19442483.mp3	Zeven plus vier is elf.	2	0				nl	
21	b51386637c01eace0c2291e27896d7	common_voice_nl_24722260.mp3	Het verslag staat echter bol van de neoliberale beleidsvoorste	2	0				nl	
22	bd3c4d0d793087205cd0adeff921bd1	common_voice_nl_20177910.mp3	Ik denk dat ik gevolgd word.	2	1				nl	
23	bf115c7dcf62b2ac4dcc49baa4d035a	common_voice_nl_24733969.mp3	Wij verzetten ons tegen deze begrotingsposten.	2	0	fifties	female		nl	
24	bf9594e371b2f199760403701a1769	common_voice_nl_17703134.mp3	Ze hadden hun gevel bordeauxrood geschilderd.	2	1				nl	
25	c18bab9891bf6f1d34a29d6e0dbddb3	common_voice_nl_17700297.mp3	Hij ligt met zijn armen gekruist op de zetel.	2	0				nl	
26	c80d131260865d629ef044c0f4a25bc	common_voice_nl_29432749.mp3	Ook de afgelopen dagen verloren weer tientallen mensen het	2	0				nl	
27	d3441ac81d7899453a061a53222cf1	common_voice_nl_18294089.mp3	De bomen worden eerst gekapt, worden dan in de zagerij tot p	2	1				nl	
28	dcdb0f48072b0c136ddc02a09429e1	common_voice_nl_22499172.mp3	Het is dus een minimumconsensus.	2	0				nl	

Looking into the data set metadata

	A	B	C	D	E	F	G	H	I	J
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3	0fca93407be6d482019f2463e60fbaf	common_voice_nl_17694848.mp3	Mijn toetsenbord zit		1				nl	
4	175d4117110538cc68a8a0157a7f0a	common_voice_nl_18441136.mp3	Ze had de bank besch		0				nl	
5	29ecd9ca5970fd1bc1297bb4a9a5f3a	common_voice_nl_30205425.mp3	Deze top heeft niet i		0					
6	3c448c76f72ee15009da6c9e474dc9f	common_voice_nl_19966492.mp3	Waar laat jij je onder		0	thirties	male		nl	
7	452cfa9d7e49e72eac2bb7ff3d5f79c	common_voice_nl_17700547.mp3	Na het lezen van vele		1	fifties	male	Nederland	nl	
8	4ac8a9187cfd11da401d6829199c18	common_voice_nl_21571581.mp3	De tampons zijn op.		0	twenties	male	Belgisch N	nl	
9	4d6898ec762a9590a973dc27579b93	common_voice_nl_25310936.mp3	Proberen wij kooldio		0				nl	
10	581251914ca5d3eeb5de368076aecc	common_voice_nl_21431804.mp3	Marijke kent Olivier r		0	twenties	female	Nederland	nl	
11	5f1e2fabc5a5c63d880ce655697814a	common_voice_nl_28458010.mp3	Mijn partij vindt dat c		0				nl	
12	723a65c399699beeb3c730e141775f	common_voice_nl_19031796.mp3	Het voeren van broo		0				nl	
13	77fc542b71acf6510c80e9a681c128f	common_voice_nl_30125083.mp3	Juni dit jaar is veel te laat.	2	0	seventies	male		nl	
14	79d570bd61fb0fcf4d363d0a1e044cf	common_voice_nl_17695742.mp3	Parket moet je stofzuigen, tegels moet je dweilen.	2	1				nl	
15	7aa3af7141d503c7c449404c3f7939	common_voice_nl_19691757.mp3	In onze buurt kent iedereen elkaar.	2	1	teens	male	Nederland	nl	
16	8d9fece2b05d9fe37e9c66e45767cae	common_voice_nl_22759350.mp3	Naar mijn oordeel valt er namelijk niet meer zoveel te verbete	2	0				nl	
17	9062e9bc9c6ee752e6f836d4296ee3	common_voice_nl_30057990.mp3	Daarom wordt voorgesteld dit soort pogingen te verbieden en	2	0				nl	
18	a01a8a30db997795dc07295a000bbc	common_voice_nl_17706390.mp3	Heb je de tafel gedekt?	2	1				nl	
19	a64ddf48b5b3b6700b146de59126a0	common_voice_nl_17696892.mp3	Ine lag in de strandstoel.	2	1				nl	
20	a9f1622fcaedf8cc1fba8d13c58564b	common_voice_nl_19442483.mp3	Zeven plus vier is elf.	2	0				nl	
21	b51386637c01eace0c2291e27896d7	common_voice_nl_24722260.mp3	Het verslag staat echter bol van de neoliberale beleidsvoorste	2	0				nl	
22	bd3c4d0d793087205cd0adeff921bd1	common_voice_nl_20177910.mp3	Ik denk dat ik gevolgd word.	2	1				nl	
23	bf115c7dcf62b2ac4dcc49baa4d035a	common_voice_nl_24733969.mp3	Wij verzetten ons tegen deze begrotingsposten.	2	0	fifties	female		nl	
24	bf9594e371b2f199760403701a1769	common_voice_nl_17703134.mp3	Ze hadden hun gevel bordeauxrood geschilderd.	2	1				nl	
25	c18bab9891bf6f1d34a29d6e0dbddb3	common_voice_nl_17700297.mp3	Hij ligt met zijn armen gekruist op de zetel.	2	0				nl	
26	c80d131260865d629ef044c0f4a25bc	common_voice_nl_29432749.mp3	Ook de afgelopen dagen verloren weer tientallen mensen het	2	0				nl	
27	d3441ac81d7899453a061a53222cf1	common_voice_nl_18294089.mp3	De bomen worden eerst gekapt, worden dan in de zagerij tot p	2	1				nl	
28	dcdb0f48072b0c136ddc02a09429e1	common_voice_nl_22499172.mp3	Het is dus een minimumconsensus.	2	0				nl	

Are all relevant metadata present?
How can you fill the missing data?

Looking into the data set metadata

	A	B	C	D	E	F	G	H	I	J
1	client_id	path	sentence	up_votes	down_votes	age	gender	accents	locale	segment
2	0d709133bf209da7f0164653b6e5f9a	common_voice_nl_17699535.mp3	De Aboriginals zijn de oorspronkelijke bewoners van AustraliÃ	2	0	fifties	male	Nederland	nl	
3	0fca93407be6d482019f2463e60fbaf	common_voice_nl_17694848.mp3	Mijn toetsenbord zit vol stof.	2	1				nl	
4	175d4117110538cc68a8a0157a7f0a	common_voice_nl_18441136.mp3	Ze had de bank beschadigd met haar skateboard.	2	0				nl	
5	29ecd9ca5970fd1bc1297bb4a9a5f3a	common_voice_nl_30205425.mp3	Deze top heeft niet in alle opzichten aan de verwachtingen vo	2	0				nl	
6	3c448c76f72ee15009da6c9e474dc9f	common_voice_nl_19966492.mp3	Waar laat jij je onderhoud doen?	2	0	thirties	male		nl	
7	452cfa9d7e49e72eac2bb7ff3d5f79c	common_voice_nl_17700547.mp3	Na het lezen van vele beoordelingen had ze eindelijk haar oog	2	1	fifties	male	Nederland	nl	
8	4ac8a9187cfd11da401d6829199c18	common_voice_nl_21571581.mp3	De tampons zijn op.	2	0	twenties	male	Belgisch N	nl	
9	4d6898ec762a9590a973dc27579b93	common_voice_nl_25310936.mp3	Proberen wij kooldioxide-emissies te beperken of proberen wi	2	0				nl	
10	581251914ca5d3eeb5de368076aecc	common_voice_nl_21431804.mp3	Marijke kent Olivier nu al meer dan twee jaar.	2	0	twenties	female	Nederland	nl	
11	5f1e2fabc5a5c63d880ce655697814a	common_v	k.	2	0				nl	
12	723a65c399699beeb3c730e141775f	common_v	is eigenlijk ongezond voor c	2	0				nl	
13	77fc542b71acf6510c80e9a681c128f	common_v		2	0	seventies	male		nl	
14	79d570bd61fb0fcf4d363d0a1e044cf	common_v	oet je dweilen.	2	1				nl	
15	7aa3af7141d503c7c449404c3f7939	common_v		2	1	teens	male	Nederland	nl	
16	8d9fece2b05d9fe37e9c66e45767cae	common_v	niet meer zoveel te verbete	2	0				nl	
17	9062e9bc9c6ee752e6f836d4296ee3	common_v	rt pogingen te verbieden en	2	0				nl	
18	a01a8a30db997795dc07295a000bbc	common_v		2	1				nl	
19	a64ddf48b5b3b6700b146de59126a0	common_v		2	1				nl	
20	a9f1622fcaedf8cc1fba8d13c58564b	common_v		2	0				nl	
21	b51386637c01eace0c2291e27896d7	common_v	neoliberale beleidsvoorste	2	0				nl	
22	bd3c4d0d793087205cd0adeff921bd1	common_v		2	1				nl	
23	bf115c7dcf62b2ac4dcc49baa4d035a	common_voice_nl_24733969.mp3	Wij verzetten ons tegen deze begrotingsposten.	2	0	fifties	female		nl	
24	bf9594e371b2f199760403701a1769	common_voice_nl_17703134.mp3	Ze hadden hun gevel bordeauxrood geschilderd.	2	1				nl	
25	c18bab9891bf6f1d34a29d6e0dbddb	common_voice_nl_17700297.mp3	Hij ligt met zijn armen gekruist op de zetel.	2	0				nl	
26	c80d131260865d629ef044c0f4a25b	common_voice_nl_29432749.mp3	Ook de afgelopen dagen verloren weer tientallen mensen het	2	0				nl	
27	d3441ac81d7899453a061a53222cf1	common_voice_nl_18294089.mp3	De bomen worden eerst gekapt, worden dan in de zagerij tot p	2	1				nl	
28	dcdb0f48072b0c136ddc02a09429e1	common_voice_nl_22499172.mp3	Het is dus een minimumconsensus.	2	0				nl	

Individual identifier

=

Not anonymous

Beware of linking risk

Looking into the data set metadata

	A	B	C	D	E	F	G	H	I	J
1	client_id	path	sentence	up_votes	down_votes	age	gender	accents	locale	segment
2	0d709133bf209da7f0164653b6e5f9a	common_voice_nl_17699535.mp3	De Aboriginals zijn de oorspronkelijke bewoners van Aust	2	0	fifties	male	Nederland	nl	
3	0fca93407be6d482019f2463e60fbaf	common_voice_nl_17694848.mp3	Mijn toetsenbord zit vol stof.	2	1				nl	
4	175d4117110538cc68a8a0157a7f0a	common_voice_nl_18441136.mp3	Ze had de bank beschadigd met haar skateboard.	2	0				nl	
5	29ecd9ca5970fd1bc1297bb4a9a5f3a	common_voice_nl_30205425.mp3	Deze top heeft niet in alle opzichten aan de verwachtingen vo	2	0				nl	
6	3c448c76f72ee15009da6c9e474dc9f	common_voice_nl_19966492.mp3	Waar laat jij je onderhoud doen?	2	0	thirties	male		nl	
7	452cfa9d7e49e72eac2bb7ff3d5f79c	common_voice_nl_17700547.mp3	Na het lezen van vele beoordelingen had ze eindelijk haar oog	2	1	fifties	male	Nederland	nl	
8	4ac8a9187cfd11da401d6829199c18	common_voice_nl_21571581.mp3	De tampons zijn op.	2	0	twenties	male	Belgisch N	nl	
9	4d6898ec762a9590a973dc27579b93	common_voice_nl_25310936.mp3	Proberen wij kooldioxide-emissies te beperken of proberen wi	2	0				nl	
10	581251914ca5d3eeb5de368076aecc	common_voice_nl_21431804.mp3	Marijke kent Olivier nu al meer dan twee jaar.	2	0	twenties	female	Nederland	nl	
11	5f1e2fabc5a5c63d880ce655697814a	common_voice_nl_28458010.mp3	Mijn partij vindt dat onverkwikkelijk.	2	0				nl	
12	723a65c399699beeb3c730e141775f	common_voice_nl_1	zond voor d	2	0				nl	
13	77fc542b71acf6510c80e9a681c128f	common_voice_nl_3		2	0	seventies	male		nl	
14	79d570bd61fb0fcf4d363d0a1e044cf	common_voice_nl_1		2	1				nl	
15	7aa3af7141d503c7c449404c3f7939	common_voice_nl_1		2	1	teens	male	Nederland	nl	
16	8d9fece2b05d9fe37e9c66e45767cae	common_voice_nl_2	l te verbete	2	0				nl	
17	9062e9bc9c6ee752e6f836d4296ee3	common_voice_nl_3	erbieden en	2	0				nl	
18	a01a8a30db997795dc07295a000bbc	common_voice_nl_1		2	1				nl	
19	a64ddf48b5b3b6700b146de59126a0	common_voice_nl_1		2	1				nl	
20	a9f1622fcaedf8cc1fba8d13c58564b	common_voice_nl_1		2	0				nl	
21	b51386637c01eace0c2291e27896d7	common_voice_nl_2	eidsvoorste	2	0				nl	
22	bd3c4d0d793087205cd0adeff921bd1	common_voice_nl_2		2	1				nl	
23	bf115c7dcf62b2ac4dcc49baa4d035a	common_voice_nl_2		2	0	fifties	female		nl	
24	bf9594e371b2f199760403701a1769	common_voice_nl_1		2	1				nl	
25	c18bab9891bf6f1d34a29d6e0dbddb3	common_voice_nl_17700297.mp3	Hij ligt met zijn armen gekruist op de zetel.	2	0				nl	
26	c80d131260865d629ef044c0f4a25bc	common_voice_nl_29432749.mp3	Ook de afgelopen dagen verloren weer tientallen mense het	2	0				nl	
27	d3441ac81d7899453a061a53222cf1	common_voice_nl_18294089.mp3	De bomen worden eerst gekapt, worden dan in de zagerij tot p	2	1				nl	
28	dcdb0f48072b0c136ddc02a09429e1	common_voice_nl_22499172.mp3	Het is dus een minimumconsensus.	2	0				nl	

How is validation performed?
Does it match the intended use?
Is it fair or could it lead to biases?
Should I provide an alternative validation?

Looking into the data set metadata

	A	B	C	D	E	F	G	H	I	J
1	client_id	path	sentence	up_votes	down_votes	age	gender	accents	ocale	segment
2	0d709133bf209da7f0164653b6e5f9a	common_voice_nl_17699535.mp3	De Aboriginals zijn de oorspronkelijke bewoners van AustraliÃ	2	0	fifties	male	Nederland	nl	
3	0fca93407be6d482019f2463e60fbaf	common_voice_nl_17694848.mp3	Mijn toetsenbord zit vol stof.	2	1				nl	
4	175d4117110538cc68a8a0157a7f0a	common_voice_nl_18441136.mp3	Ze had de bank beschadigd met haar skateboard.	2	0				nl	
5	29ecd9ca5970fd1bc1297bb4a9a5f3a	common_voice_nl_30205425.mp3	Deze top heeft niet in alle opzichten aan de verwachtingen vo	2	0				nl	
6	3c448c76f72ee15009da6c9e474dc9f	common_voice_nl_19966492.mp3	Waar laat jij je onderhoud doen?	2	0	thirties	male		nl	
7	452cfa9d7e49e72eac2bb7ff3d5f79c	common_voice_nl_17700547.mp3	Na het lezen van vele beoordelingen had ze eindelijk haar oog	2	1	fifties	male	Nederland	nl	
8	4ac8a9187cfd11da401d6829199c18	common_voice_nl_21571581.mp3	De tampons zijn op.	2	0	twenties	male	Belgisch N	nl	
9	4d6898ec762a9590a973dc27579b93	common_voice_nl_25310936.mp3	Proberen wij kooldioxide-emissies te beperken of proberen wi	2	0				nl	
10	581251914ca5d3eeb5de368076aecc	common_voice_nl_21431804.mp3	Marijke kent Olivier nu al meer dan twee jaar.	2	0	twenties	femal	Nederland	nl	
11	5f1e2fabc5a5c63d880ce655697814a	common_voice_nl_28458010.mp3	Mijn partij vindt dat onverkwik						nl	
12	723a65c399699beeb3c730e141775f	common_voice_nl_19031796.mp3	Het voeren van brood aan eer						nl	
13	77fc542b71acf6510c80e9a681c128f	common_voice_nl_30125083.mp3	Juni dit jaar is veel te laat.				male		nl	
14	79d570bd61fb0fcf4d363d0a1e044cf	common_voice_nl_17695742.mp3	Parket moet je stofzuigen, teg						nl	
15	7aa3af7141d503c7c449404c3f7939	common_voice_nl_19691757.mp3	In onze buurt kent iedereen el				mal	Nederland	nl	
16	8d9fece2b05d9fe37e9c66e45767cae	common_voice_nl_22759350.mp3	Naar mijn oordeel valt er nam						nl	
17	9062e9bc9c6ee752e6f836d4296ee3	common_voice_nl_30057990.mp3	Daarom wordt voorgesteld dit						nl	
18	a01a8a30db997795dc07295a000bbc	common_voice_nl_17706390.mp3	Heb je de tafel gedekt?						nl	
19	a64ddf48b5b3b6700b146de59126a0	common_voice_nl_17696892.mp3	Ine lag in de strandstoel.						nl	
20	a9f1622fcaedf8cc1fba8d13c58564b	common_voice_nl_19442483.mp3	Zeven plus vier is elf.						nl	
21	b51386637c01eace0c2291e27896d7	common_voice_nl_24722260.mp3	Het verslag staat echter bol van de neoliberale beleidsvoorste	2	0				nl	
22	bd3c4d0d793087205cd0adeff921bd1	common_voice_nl_20177910.mp3	Ik denk dat ik gevolgd word.	2	1				nl	
23	bf115c7dcf62b2ac4dcc49baa4d035a	common_voice_nl_24733969.mp3	Wij verzetten ons tegen deze begrotingsposten.	2	0	fifties	femal		nl	
24	bf9594e371b2f199760403701a1769	common_voice_nl_17703134.mp3	Ze hadden hun gevel bordeauxrood geschilderd.	2	1				nl	
25	c18bab9891bf6f1d34a29d6e0dbddb3	common_voice_nl_17700297.mp3	Hij ligt met zijn armen gekruist op de zetel.	2	0				nl	
26	c80d131260865d629ef044c0f4a25bc	common_voice_nl_29432749.mp3	Ook de afgelopen dagen verloren weer tientallen mensen het	2	0				nl	
27	d3441ac81d7899453a061a53222cf1	common_voice_nl_18294089.mp3	De bomen worden eerst gekapt, worden dan in de zagerij tot p	2	1				nl	
28	dcdb0f48072b0c136ddc02a09429e1	common_voice_nl_22499172.mp3	Het is dus een minimumconsensus.	2	0				nl	

What accents are present?
Are foreigners and vulnerable groups represented?

Looking into the data set metadata

	A	B	C	D	E	F	G	H	I	J
1	client_id	path	sentence	up_votes	down_votes	age	gender	accents	locale	segment
2	0d709133bf209da7f0164653b6e5f9a	common_voice_nl_17699535.mp3	De Aboriginals zijn de oorspronkelijke bewoners van AustraliÃ	2		0 fifties	male	Nederland	nl	
3	0fca93407be6d482019f2463e60fbaf	common_voice_nl_17694848.mp3	Mijn toetsenbord zit vol stof.	2		1			nl	
4	175d4117110538cc68a8a0157a7f0a	common_voice_nl_18441136.mp3	Ze had de bank beschadigd met haar skateboard.	2		0			nl	
5	29ecd9ca5970fd1bc1297bb4a9a5f3a	common_voice_nl_30205425.mp3	Deze top heeft niet in alle opzichten aan de verwachtingen vo	2		0			nl	
6	3c448c76f72ee15009da6c9e474dc9f	common_voice_nl_19966492.mp3	Waar laat jij je onderhoud doen?	2		0 thirties	male		nl	
7	452cfa9d7e49e72eac2bb7ff3d5f79c	common_voice_nl_17700547.mp3	Na het lezen van vele beoordelingen had ze eindelijk haar oog	2		1 fifties	male	Nederland	nl	
8	4ac8a9187cfd11da401d6829199c18	common_voice_nl_21571581.mp3	De tampons zijn op.	2		0 twenties	male	Belgisch N	nl	
9	4d6898ec762a9590a973dc27579b93	common_voice_nl_25310936.mp3	Proberen wij kooldioxide-emissies te beperken of proberen wi	2		0			nl	
10	581251914ca5d3eeb5de368076aecc	common_voice_nl_21431804.mp3	Marijke			0 twenties	female	Nederland	nl	
11	5f1e2fabc5a5c63d880ce655697814a	common_voice_nl_28458010.mp3	Mijn par			0			nl	
12	723a65c399699beeb3c730e141775f	common_voice_nl_19031796.mp3	Het voer			0			nl	
13	77fc542b71acf6510c80e9a681c128f	common_voice_nl_30125083.mp3	Juni dit ja			0 seventies	male		nl	
14	79d570bd61fb0fcf4d363d0a1e044cf	common_voice_nl_17695742.mp3	Parket m			1			nl	
15	7aa3af7141d503c7c449404c3f7939	common_voice_nl_19691757.mp3	In onze b			1 teens	male	Nederland	nl	
16	8d9fece2b05d9fe37e9c66e45767cae	common_voice_nl_22759350.mp3	Naar mij			0			nl	
17	9062e9bc9c6ee752e6f836d4296ee3	common_voice_nl_30057990.mp3	Daarom			0			nl	
18	a01a8a30db997795dc07295a000bbc	common_voice_nl_17706390.mp3	Heb je de			1			nl	
19	a64ddf48b5b3b6700b146de59126a0	common_voice_nl_17696892.mp3	Ine lag in			1			nl	
20	a9f1622fcaedf8cc1fba8d13c58564b	common_voice_nl_19442483.mp3	Zeven pl			0			nl	
21	b51386637c01eace0c2291e27896d7	common_voice_nl_24722260.mp3	Het verslag staat echter bol van de neoliberale beleidsvoorste			0			nl	
22	bd3c4d0d793087205cd0adeff921bd1	common_voice_nl_20177910.mp3	Ik denk dat ik gevolgd word.	2		1			nl	
23	bf115c7dcf62b2ac4dcc49baa4d035a	common_voice_nl_24733969.mp3	Wij verzetten ons tegen deze begrotingsposten.	2		0 fifties	female		nl	
24	bf9594e371b2f199760403701a1769	common_voice_nl_17703134.mp3	Ze hadden hun gevel bordeauxrood geschilderd.	2		1			nl	
25	c18bab9891bf6f1d34a29d6e0dbddb3	common_voice_nl_17700297.mp3	Hij ligt met zijn armen gekruist op de zetel.	2		0			nl	
26	c80d131260865d629ef044c0f4a25bc	common_voice_nl_29432749.mp3	Ook de afgelopen dagen verloren weer tientallen mensen het	2		0			nl	
27	d3441ac81d7899453a061a53222cf1	common_voice_nl_18294089.mp3	De bomen worden eerst gekapt, worden dan in de zagerij tot p	2		1			nl	
28	dcdb0f48072b0c136ddc02a09429e1	common_voice_nl_22499172.mp3	Het is dus een minimumconsensus.	2		0			nl	

Are different age and gender types equally represented?
Could an imbalance lead to discrimination?

TU/e

DATA SET FAIRNESS 1

What is data set fairness?

Research has shown how **data sets convey social bias** in AI systems, especially those based on machine learning. A biased data set is **not representative of reality** and might contribute to perpetuate societal biases within the model. To tackle this problem, it is important to understand how to **avoid biases, errors, and unethical practices** while **creating the data sets**.

DATA SET FAIRNESS 2

What is data set fairness?

Mainstream works in computer science addresses the problem of fairness in a context of prediction-based decisions. Usually, mathematical definitions requires **achieving parities across distributions of outcomes or scores** (common examples include statistical parity and calibration). Definitions can refer to **individuals** (e.g., requiring that similar individuals are treated similarly) or **groups** (e.g., requiring that any error measure is equal across groups, defined by so-called protected characteristics like gender or race).

Mathematical definitions can reflect philosophical conceptualisations, such as the **equality of outcomes and the equality of opportunities**. In general, computer science literature expresses in empirical and formal terms philosophical problems of **distributive justice** in which the goal of fairness is to achieve an **equal allocation of goods**.

DATA SET FAIRNESS 3

What is data set fairness?

The concept of fairness is frequently found also in **text of law** and it has been extensively analysed by the legal scholarship, having its roots in the **"bona fide" concept in Roman law**. The fairness principle is an **overarching obligation** in the European Digital Strategy, for example in Article 5, par. 1, of GDPR, and in the Data Act, but it is also found in the EU and in other countries in many pieces of law, such as Article 8, par. 2, of the Charter of Fundamental Rights of the European Union, contract law, and others.

The data protection law scholarship divide the fairness principle into two elements, **procedural fairness** and **fair balancing**, noting that the concept of fairness is related to the idea of **vulnerability**. The GDPR aims at protecting data subjects, who are in a vulnerable position as opposed to data controllers, therefore it tries to make a balance between the fundamental rights and freedoms of the former and the interests of the latter.

FAIRNESS ASSESSMENT OF DATA SETS

So far, the assessment of algorithmic fairness in the context of data has followed **two main directions**. A research track regards the analysis of **data curation processes** or the development of new **documentation frameworks**. Recently, Fabris et. Al annotated hundreds of data sets employed in algorithmic research efforts specifying key fairness-related characteristics (e.g. the domain of application, the target task and the sensitive attributes encoded in the data set) in order to address the so-called “documentation debt”.

Another direction focuses on the **empirical analysis of fairness-related data dimensions**. These efforts include **pre-processing techniques** such as sanitization and instance re-weighting and procedures evaluating distinct sources of discrimination in data collection or studying the effect of data-related factors on the fairness of algorithms.

OUR WORK

Investigating Semi-Automatic Assessment of Data-set Fairness by Means of Fuzzy Logic

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FANFAIR AIMS

Our work moves along the line of empirical research with some important distinctions. The fairness-related features we explore are **agnostic about the sensitive attribute** in that they capture essential vulnerabilities that may contribute to model (un)fairness.

Another distinct aspect of our approach is to integrate **statistical information** about the data sets (**balance, numerosity, unevenness, incompleteness**) with **qualitative considerations** (**compliance, quality**) which take into account legal and ethical requirements of data-related practices (e.g. GDPR, principles of research integrity). This integration is operated through the application of **fuzzy rule-base logic** and reflected in a score that can be indicative of the fairness potentially introduced by a given data set.

FANFAIR DEFINITION OF FAIRNESS

Although in our paper we take into consideration GDPR, we do not limit the concept of (legal) fairness to data protection law, but we expand the notion to other areas of law, including the **non-discrimination principle**, that is found, inter alia, in the European Convention of Human Rights, the **equity principle**, and the **bona fide principle** in contractual agreements.

In our opinion, a data set which data collection has been flawed by discrimination or another breach of the law (e.g., copyright law, criminal law, medical law) cannot be considered fair.

The **lawfulness criteria** is also in line with the High-Level Expert Group on AI (HLEG AI)'s guidelines on Trustworthy AI.

FANFAIR FUZZY LOGIC ASSESSMENT

In order to assess the potential fairness of a data set, we identified six different elements that should be addressed before employing the data set. To identify each criteria, we were inspired by general principle of civil law and EU law, and established ethics guidelines.

TABLE I
DESCRIPTION OF THE DATA SET FEATURES / LINGUISTIC VARIABLES CONSIDERED IN THIS WORK.

Feature	Meaning
balance	How balanced is the data set, with respect to the output labels.
numerosity	The numerosity of instances with respect to the number of features.
evenness	A metric assessing the amount of outliers in the data set.
compliance	How much the data set is compliant with respect to governmental rules.
quality	A user-defined evaluation of the overall quality of the data set.
incompleteness	This metric measures the amount of missing data in the data set.

BALANCE

A data set can be defined as "balanced" if **all classes are well represented** in the data set. The importance of balance is described in a wealth of literature: an unbalanced data set may produce biased models, unable to generalise and with prejudicial effects on single individuals or entire categories of individuals, such as disparity in accuracy.

For example, if a data set, which is used to train a model aimed at classifying patients in healthy or non-healthy, contains **thousands of examples of non-healthy patients** and only a **few example of healthy individuals**, the model will probably be unable to recognise healthy individuals with high accuracy. It is therefore crucial to have a similar number of examples in each class.

NUMEROSITY

This data set feature refers to the **quantity of instances** normalised by the **number of features** in the data set. The quantity is important in order to mitigate possible biases in the model output: a model with a low number of examples will not be able to generalise.

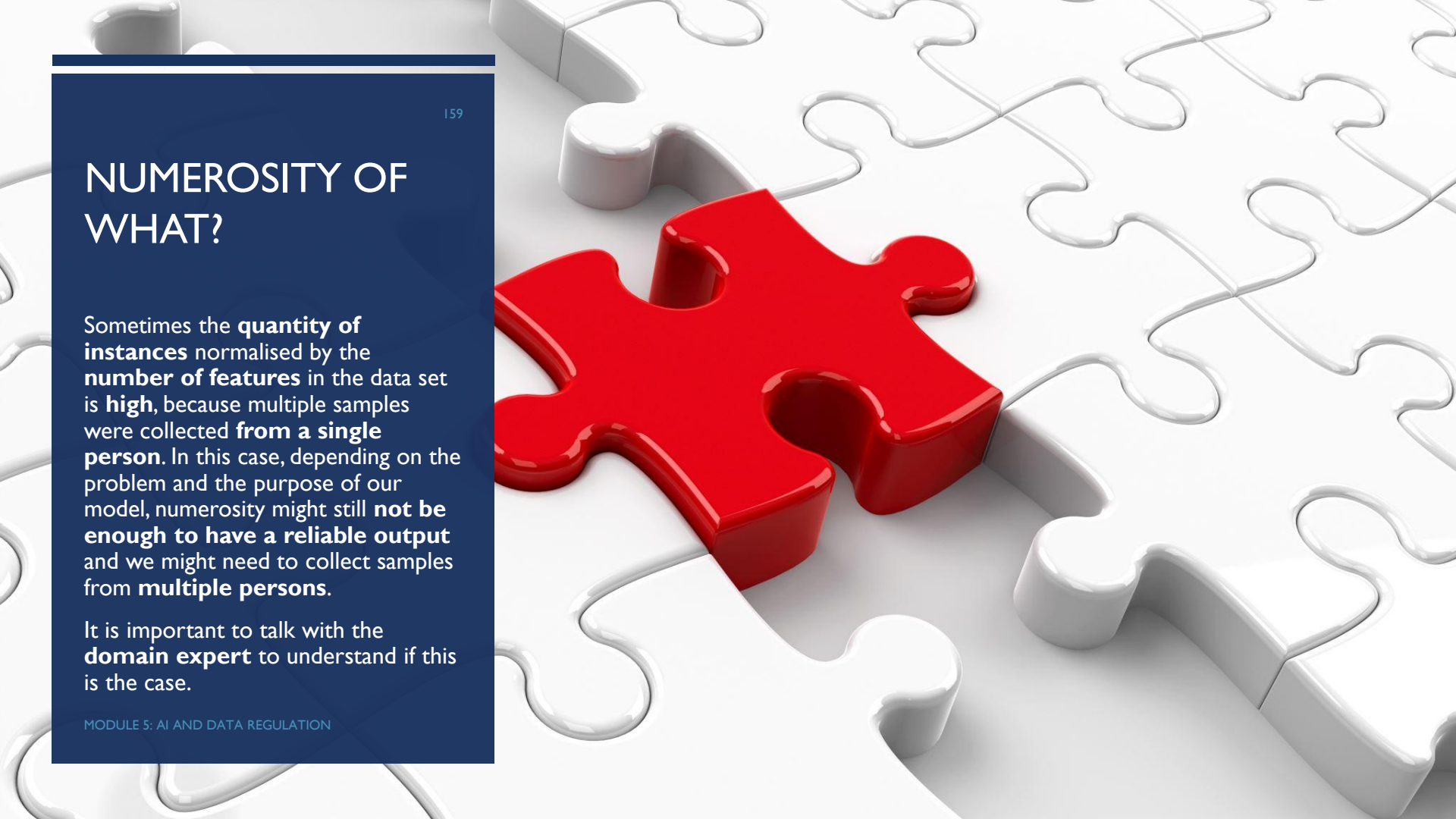
To show an example of why this can happen, consider the situation in which all the features have no relation with the outputs to be predicted. In that case, we would like to have a machine learning algorithm to be unable to perform well since there is actually no relation with the between the inputs and the outputs: any learned relation will be purely the effect of chance, it will not be representative of the phenomenon being modelled and will clearly be unable to generalise.

A widespread **rule of thumb** of machine learning suggests that an optimal number of instances S would be 10 × D.

NUMEROSITY OF WHAT?

Sometimes the **quantity of instances** normalised by the **number of features** in the data set is **high**, because multiple samples were collected **from a single person**. In this case, depending on the problem and the purpose of our model, numerosity might still **not be enough to have a reliable output** and we might need to collect samples from **multiple persons**.

It is important to talk with the **domain expert** to understand if this is the case.



UNEVENNESS

This feature refers to the **number of outliers in the examples**. If a value is very different from all other values in the data set, it is important to understand why this occurs. For example, in a data set made of physiological values of non-healthy patients, in the weight column we could find abnormal values (too high or too low).

The **domain expert** will certainly understand that an adult individual cannot have a weight of 100 grams or of 20000 kilograms, and therefore there is an **error in the data collection**. However, outliers might sometimes be just **individuals with unique features** with respect to the average, such as a patient with a weight over 600 kilograms. In this case, there is a risk of singling out and the patient's identity will likely be recognized if the data set is released to the public.

It is important to note that **outliers are not useful for the model**, since they refers to particular cases which might better be assessed individually by the domain expert instead of by an automatic mean such as an AI model. The model will not be able to generalise and give an accurate output for outliers' cases. Instead, it would be more useful to create a different model specifically calibrated to individuals who present abnormal values.

QUALITY

This feature refers to the quality of data as **assessed by the domain expert** (such as if an image is clear or complete, or if the sound can be recognised in a recording). The domain expert will make an **assessment** about the quality of the data set, based on the state of the art in the specific field (such as medical imaging) and on the needs related to each specific model.

This step is needed because the data collection is often performed by different people without direct supervision (e.g., nurses, young medical trainees, students, etc.), therefore errors in acquiring clear data may occur.

Do not use a data set to train models employed in **critical applications** (e.g., clinical decisions) if a domain expert has not validated the quality of the data.

INCOMPLETENESS

This feature refers to "**holes**" in the data set regarding **relevant elements**, such as **missing information** about a patient in one or more physiological values. When this circumstance occurs, it is necessary to exploit **imputation techniques** or to **discard the patient** from the data set altogether, if too little information is provided. There are three possible types of incomplete data sets:

1) some **variables** (e.g., dataframe columns) may have **missing values**. For instance, this might happen when the variable is related to some exam that is only administered to very specific patients;

2) some **instances** (e.g., dataframe rows) may have **missing values**. For instance, this might happen to fragile patients that cannot undergo some treatments or exams;

3) finally, the holes can be **randomly distributed** across all variables and instances of the data sets.

RESULTS OF FANFAIR

These variables represent important characteristics that are automatically extracted by the data set or (in the case of compliance and quality) are entered by the user. FAnFAIR returns a summary of the contribution of all variables to the final fairness value. The contributions are expressed both in fuzzy terms (e.g., "compliance is fairly high") and by using a visual plot. The final fairness value is calculated as a value between 0 (totally unfair) and 1 (maximum theoretical fairness).

TABLE II
RULE-BASE USED BY FANFAIR FOR THE ASSESSMENT OF DATA SET FAIRNESS.

Rule 1:	IF balance IS high THEN φ IS high_fairness
Rule 2:	IF balance IS low THEN φ IS low_fairness
Rule 3:	IF numerosity IS high THEN φ IS high_fairness
Rule 4:	IF numerosity IS low THEN φ IS low_fairness
Rule 5:	IF unevenness IS high THEN φ IS low_fairness
Rule 6:	IF unevenness IS low THEN φ IS high_fairness
Rule 7:	IF compliance IS high THEN φ IS high_fairness
Rule 8:	IF compliance IS low THEN φ IS low_fairness
Rule 9:	IF quality IS high THEN φ IS high_fairness
Rule 10:	IF quality IS low THEN φ IS low_fairness
Rule 11:	IF incompleteness IS high THEN φ IS low_fairness
Rule 12:	IF incompleteness IS low THEN φ IS high_fairness

Existing techniques (pros & cons)

- Select a statistically significant sample and check it manually
- Over- or under-sampling
- Make your *testing* data set balanced
- Use automated techniques, e.g., automated gender recognition, to fill the gaps in metadata
- Use multiple data sets and combine them
- Collect new data to integrate missing data
- Use data augmentation and/or synthetic data
- Pre-process your data to add noise or other relevant elements
- Compare your model output with that of different models in different settings

DATA SET BALANCING

If there is a severe imbalance and the dataset is tiny (or high dimensional, etc.), which is frequently the situation in clinical data sets, imbalance could be problematic. Having unbalanced data can be an issue depending on the application's objectives.

For instance, if you want to forecast a given outcome and your data shows that A occurs 99.99% of the time and B occurs 0.01% of the time, your algorithm would likely always suggest A. However, in many cases, we are more curious about why B occurs on occasion than we are about whether the prediction was accurate. We might be interested in good performance on a minority class. Unbalanced data, therefore, becomes a concern. The technique you used is appropriate, but not for your question.

See: https://stats.stackexchange.com/questions/283170/when-is-unbalanced-data-really-a-problem-in-machine-learning

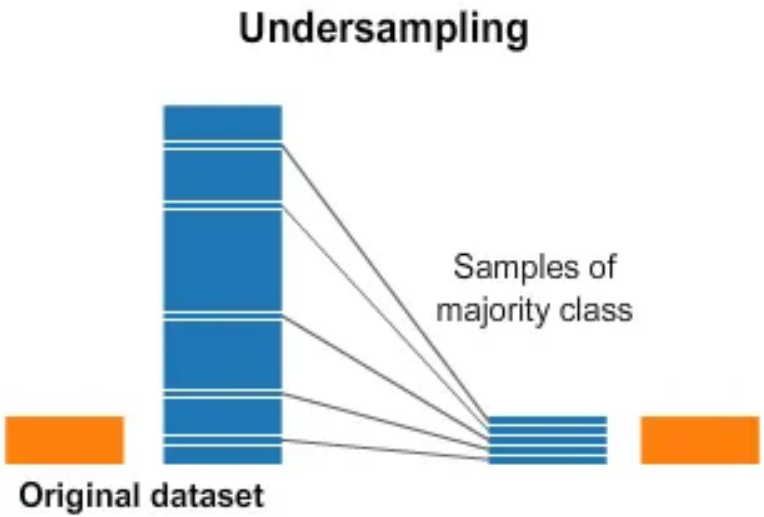
HOW DO WE SOLVE DATA SET UNBALANCE?

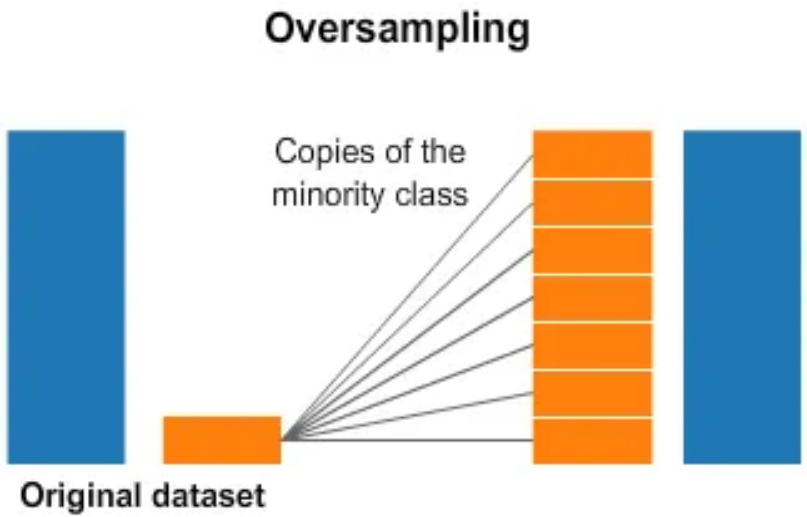
- Baseline Methods
  - Random over-sampling
  - Random under-sampling
- Under-sampling Methods
  - Tomek links
  - Condensed Nearest Neighbor Rule
  - One-sided selection
  - CNN + Tomek links
  - Neighborhood Cleaning Rule
- Over-sampling Methods
  - Smote
- Combination of Over-sampling method with Under-sampling method
  - Smote + Tomek links
  - Smote + ENN

Source: A Study of the Behavior of Several Methods for Balancing Machine Learning Training Data Author: Gustavo E. A. Batista Presenter: Hui Li University of Ottawa.

OVERSAMPLING AND UNDERSAMPLING

You shouldn't over-sample the test set, only the training test





Source: https://medium.com/towards-data-science/handling-imbalanced-datasets-in-deep-learning-f48407a0e758

DATA AUGMENTATION

It is a set of techniques that expand the available dataset without actually collecting new elements: data augmentation applies controlled random changes to existing data, making modified copies of it, in order to reduce overfitting.

Like all techniques, it can introduce biases.



IMPUTATION

Imputation techniques "guess" realistic values that are used to replace missing values (“holes”) in a dataset. The model guesses data by assuming that they are similar to the ones that already are present in the data set, which is going to reinforce any pattern that might be in the data, potentially turning a non-significant pattern into a significant one and thus introducing biases.

Example: Missing data can substantially affect the precision of estimated change in patients reported outcomes scores from clinical registry data. Inclusion of auxiliary information in Multiple Imputation models can increase precision and reduce bias, but identifying the optimal auxiliary variable(s) may be challenging.

(Ayilara, O.F., Zhang, L., Sajobi, T.T. et al. Impact of missing data on bias and precision when estimating change in patient-reported outcomes from a clinical registry. Health Qual Life Outcomes 17, 106 (2019). https://doi.org/10.1186/s12955-019-1181-2)

SYNTETHIC DATA

- Synthetic data is information that's artificially created, it is not collected or generated by real-world events or persons. Generally, it is created by algorithms, but of course it can be generated by hand as well
- Synthetic data and original data should deliver very similar results when undergoing the same statistical analysis
- It has the benefit to protect privacy, as real personal data can be avoided. A privacy assurance assessment should be performed to ensure that the resulting synthetic data is not actual personal data, evaluating the extent to which data subjects can be identified in the synthetic data and how much new data about those data subjects would be revealed upon successful identification
- It might contribute to mitigate bias by using fair synthetic datasets to train artificial intelligence models, manipulated to have a better representativeness of the world
- The quality of synthetic data is highly correlated with the quality of the original data and the data generation model. Synthetic data may reflect the biases in original data. Also, the manipulation of datasets to create fair synthetic datasets might result in inaccurate data

ANY QUESTION?

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